# CIVL4340
# Wind Engineering

Week 1: Introduction to wind engineering and its application & Meteorology of windstorms

The University Of Queensland
AUSTRALIA

# What is wind engineering?

*"Wind engineering is best defined as the rational treatment of the interactions between wind in the atmospheric boundary layer and people and their works on the surface of earth."*

*Based on quote from Prof. Jack Cermak*

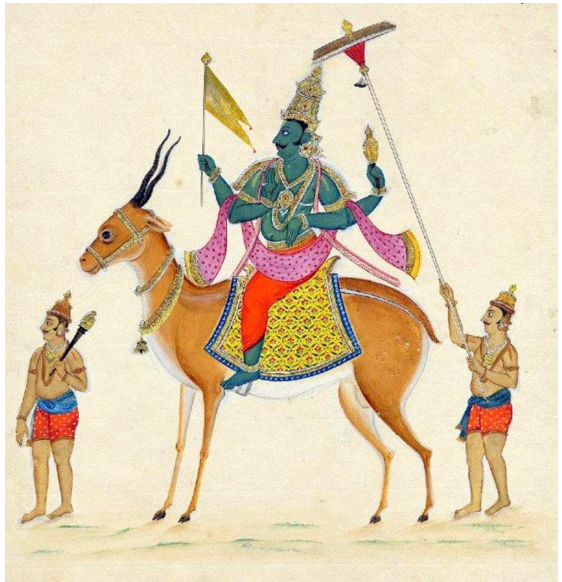
# What is wind engineering?

Wind engineering is an interdisciplinary field, drawing upon:

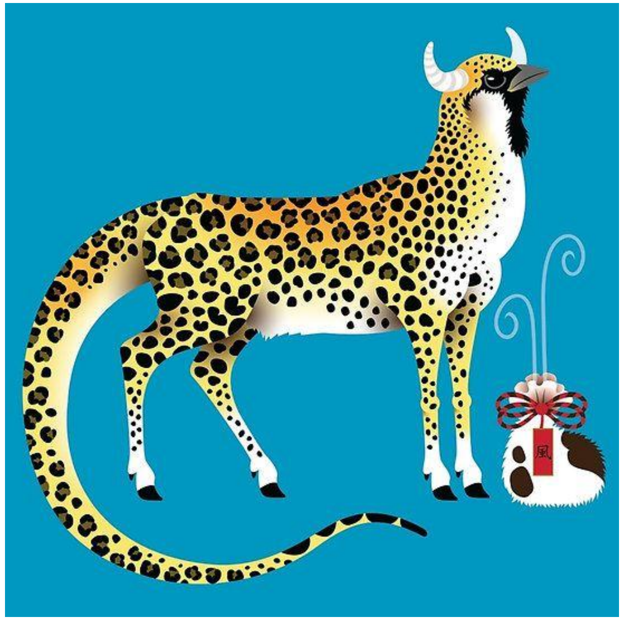
- Structural Engineering → dynamic response, load paths
- Meteorology → atmospheric boundary layers, turbulence, storm dynamics
- Aerodynamics → bluff-body aerodynamics, wind turbine aerodynamics
- Aeronautics → aerolasticity (e.g., flutter)
- Fluid Mechanics → computational fluid dynamics, Fluid-structure interactions
- Signal Processing → frequency domain and time domain analysis
- Computing → large-scale, parallel systems
- Experimentation → wind tunnels, large-scale testing
- Statistics → extreme value analysis, energy forecasting, climatology
- ….more

Anemoi, Greek mythology

Vayu, Hinduism



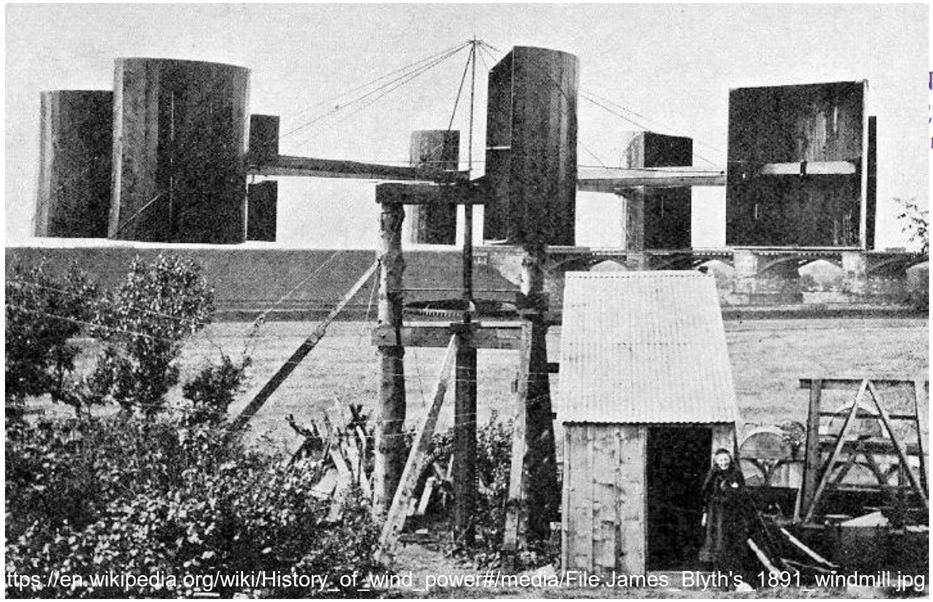
Njörðr, Viking mythology

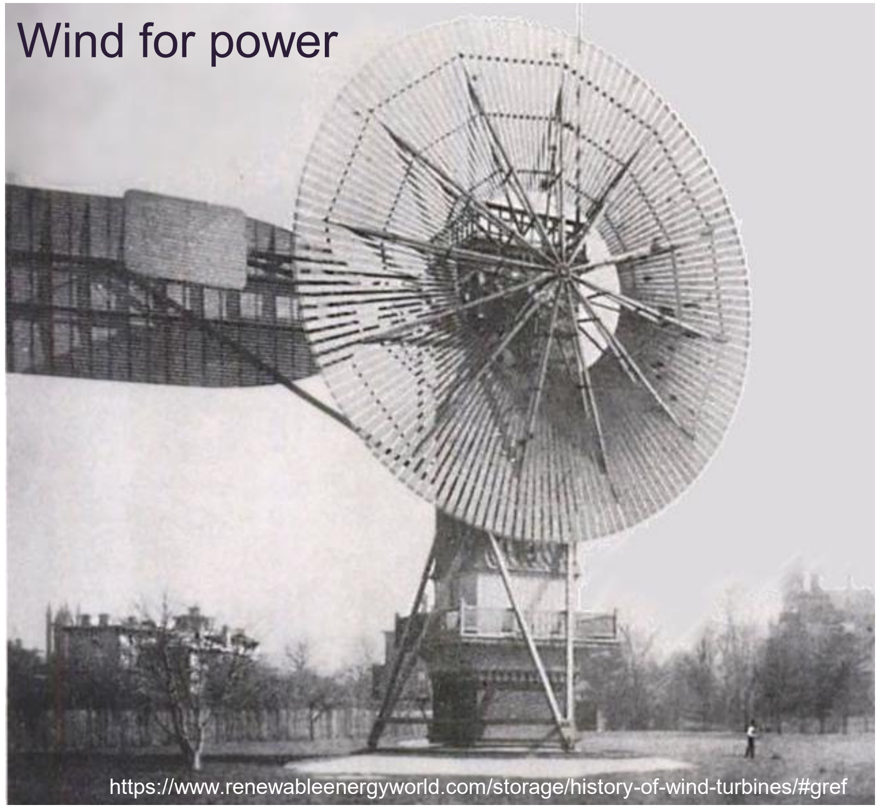


Fei Lian, Chinese wind spirit

Fujin, Japanese Shintoism


Wind for transport


ttps://en.wikipedia.org/wiki/History_of_wind_power#/media/File:James_Blyth's_1891_windmill.jpg


Wind for power
https://www.renewableenergyworld.com/storage/history-of-wind-turbines/#gref

# The Beaufort Scale

"Over thousands of years sailors have learnt to estimate the speed of the wind just by looking about. This technique matured into what we now call the Beaufort scale. The universe tells you everything you need to know about it as long as you are prepared to watch, to listen, to smell, in short to observe" .....Howtoons 2006

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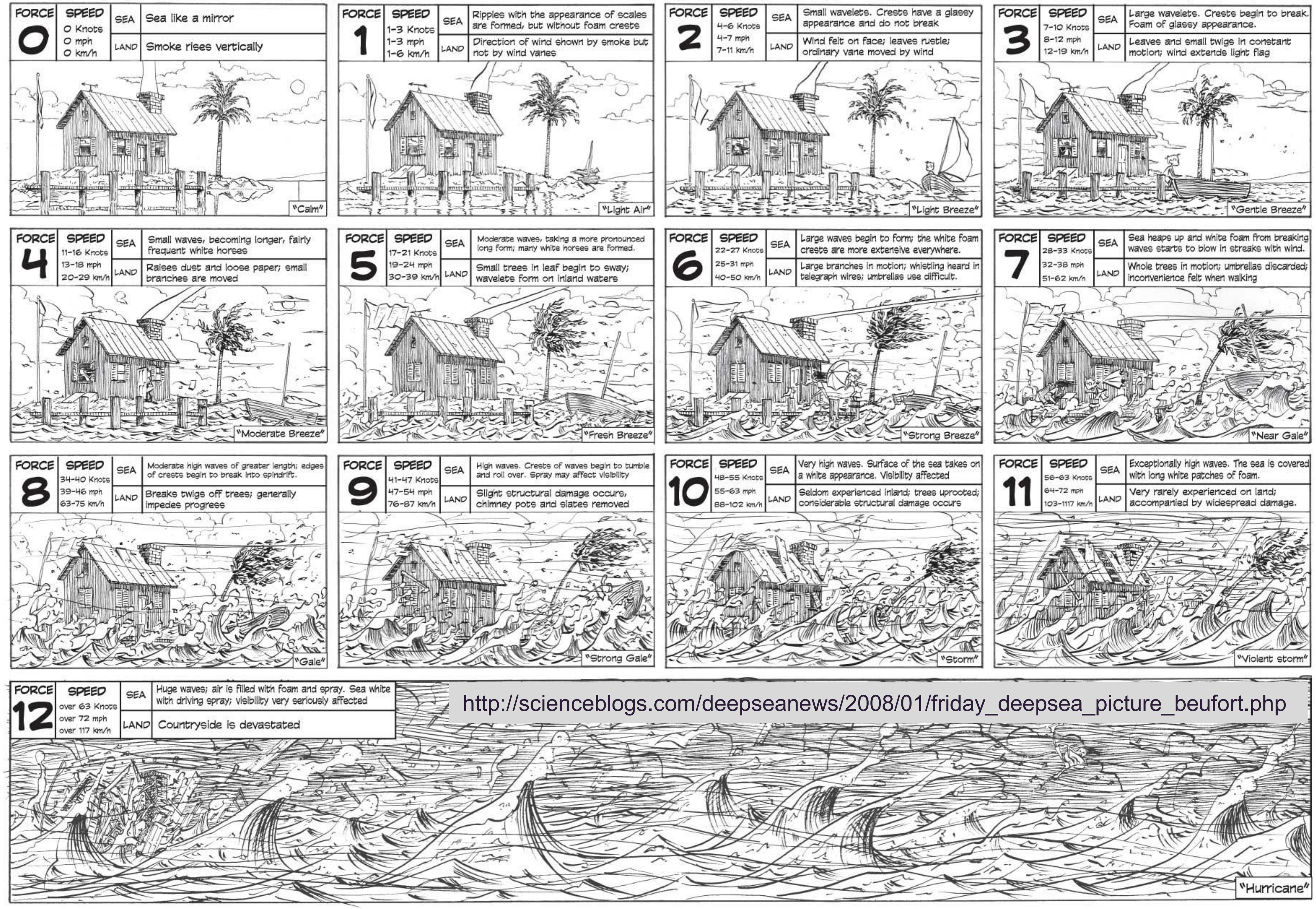
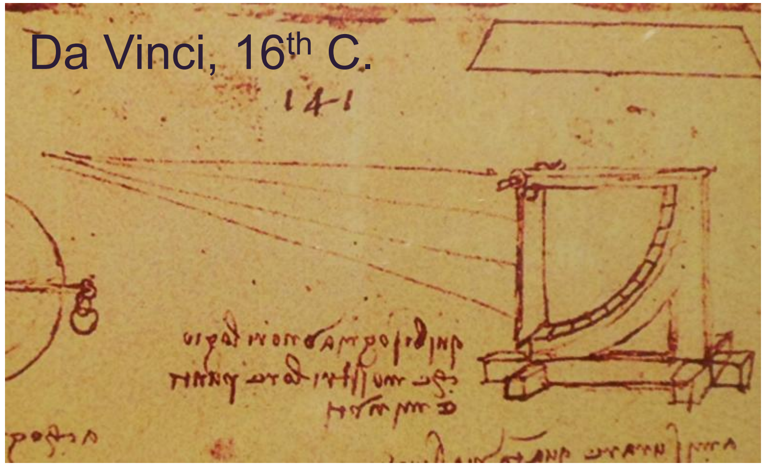
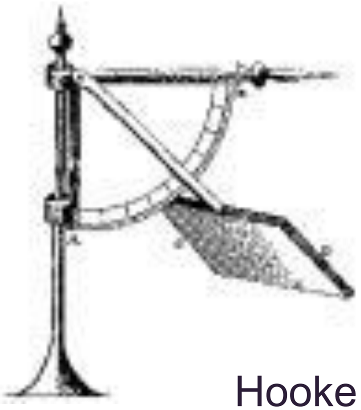



http://scienceblogs.com/deepseanews/2008/01/friday_deepsea_picture_beufort.php

# Measuring the wind



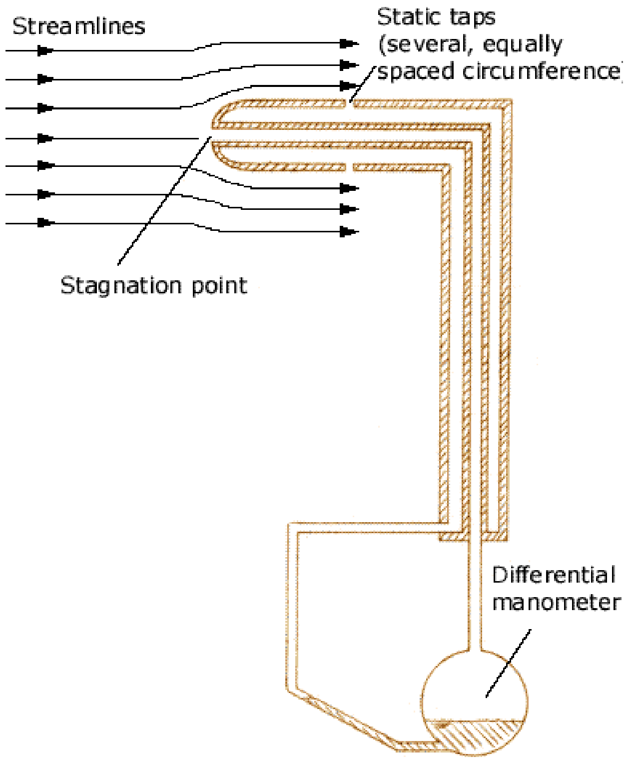
Da Vinci, 16th C.



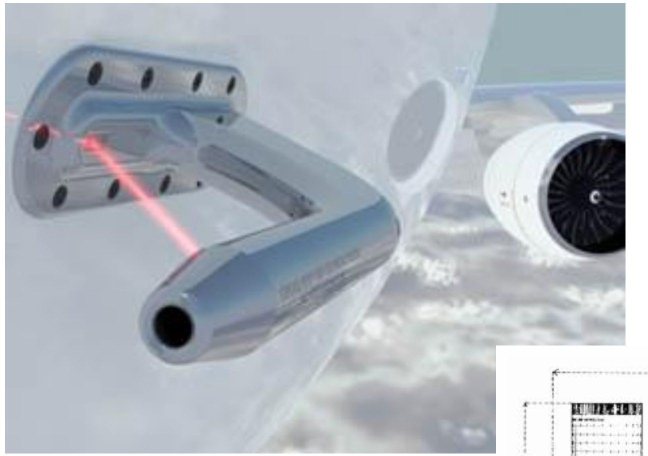
Hooke, 17th C.

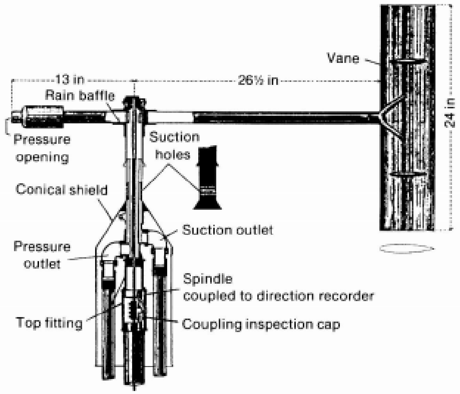
Pendulum (pressure plate) anemometers

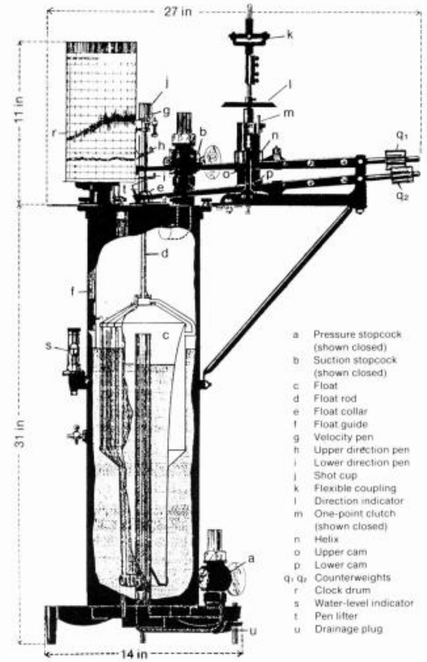
## Dynamic pressure anemometers




Huet, 1722






Dines, 1892

http://www.keyshone.com/wp-content/uploads/2014/11/pitot-tube-21.jpg

# Measuring the wind

Robinson, 1846

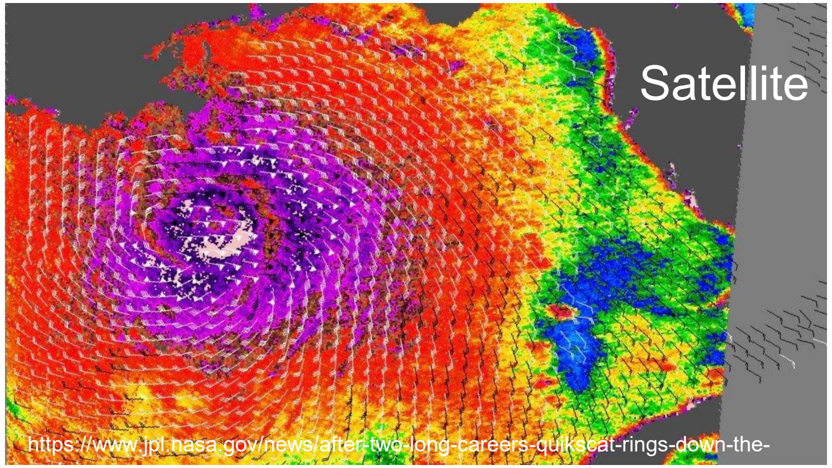
Synchrotac (BoM)

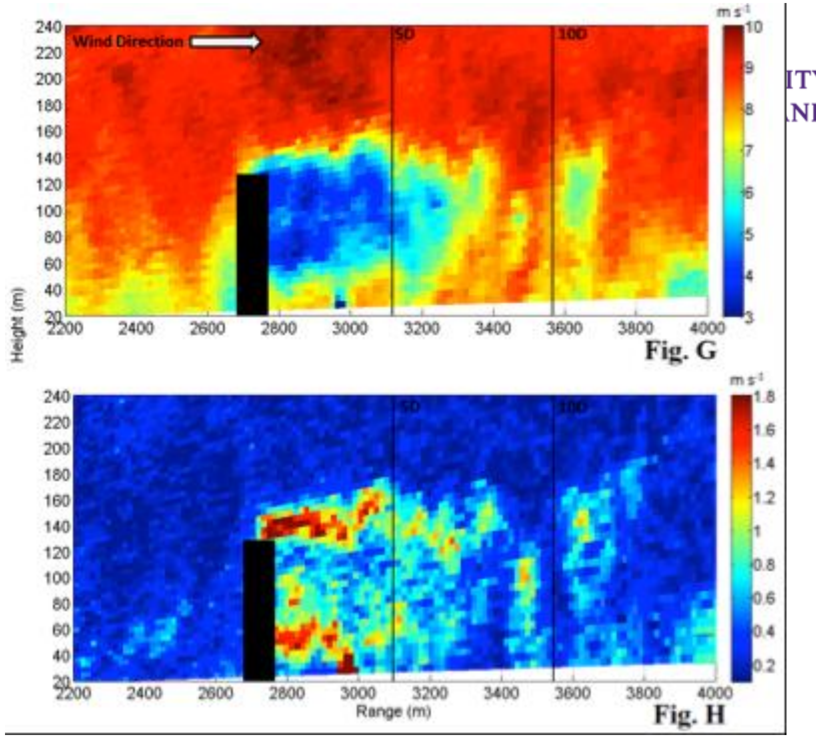

Cup anemometers

Propeller anemometers

Sonic anemometers

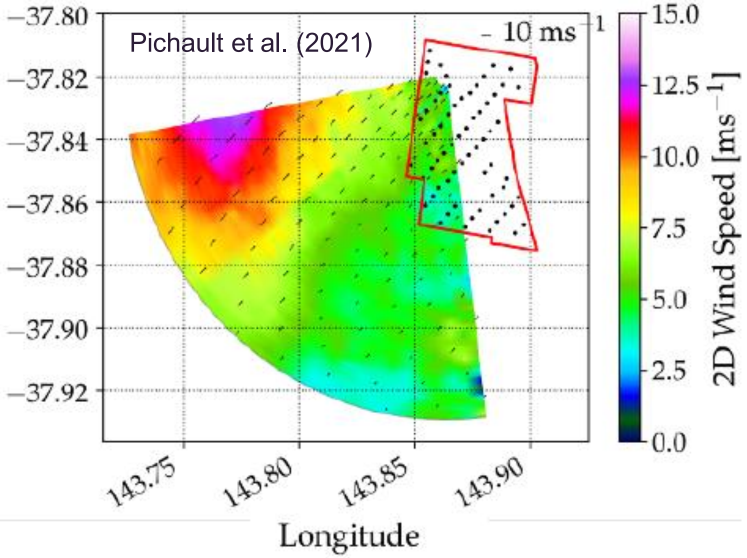
# Measuring the wind



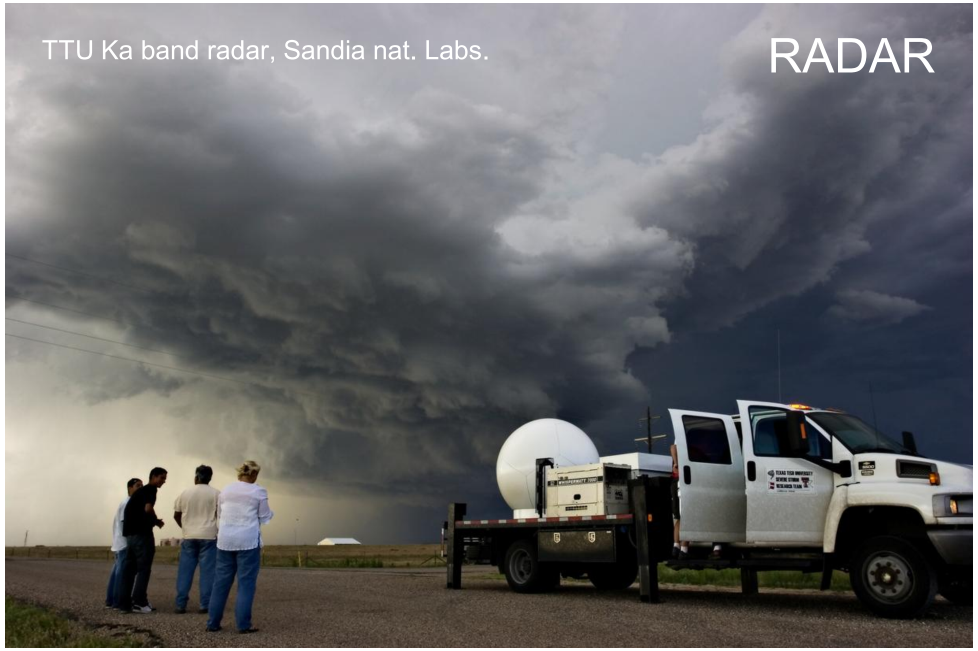





LiDAR

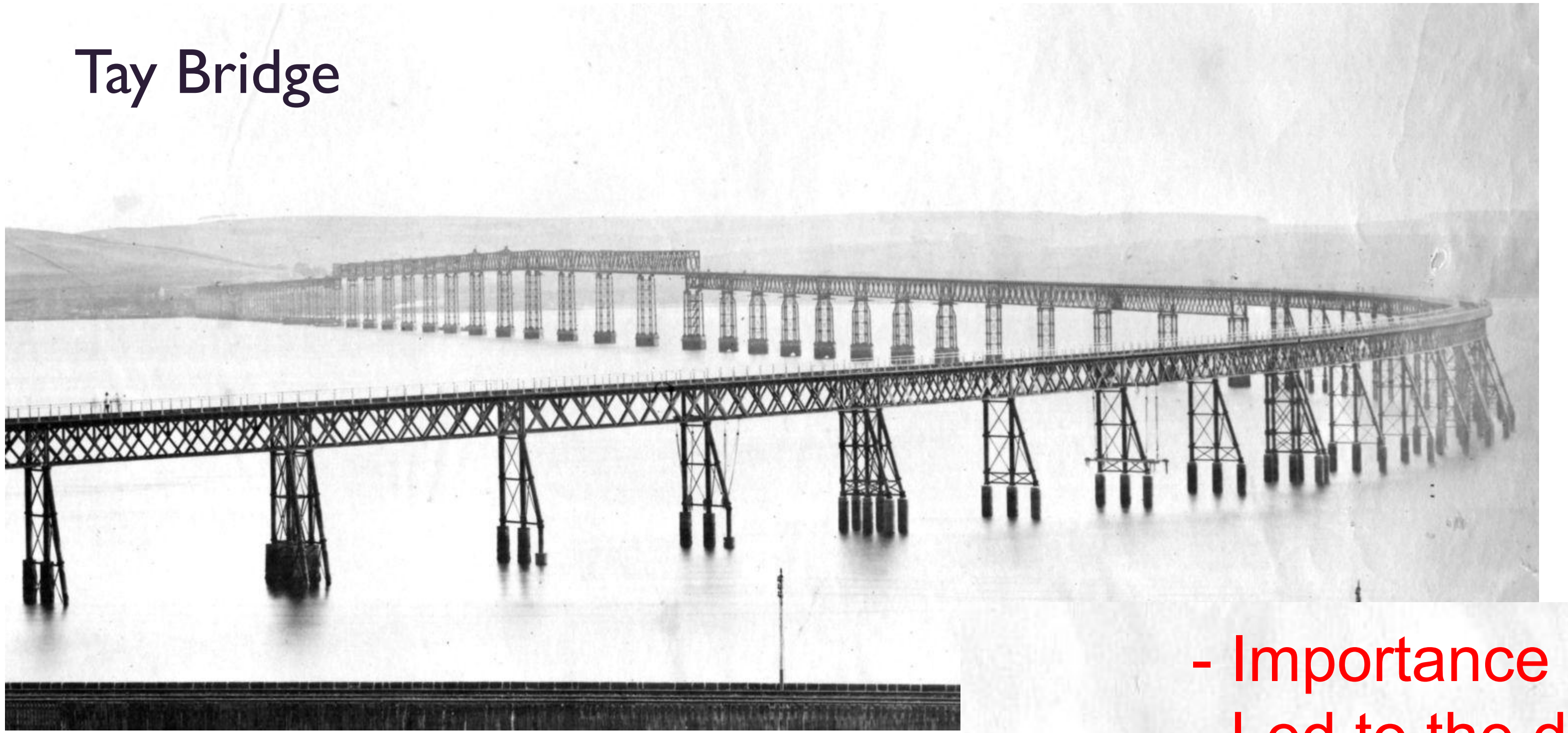



Remote sensing

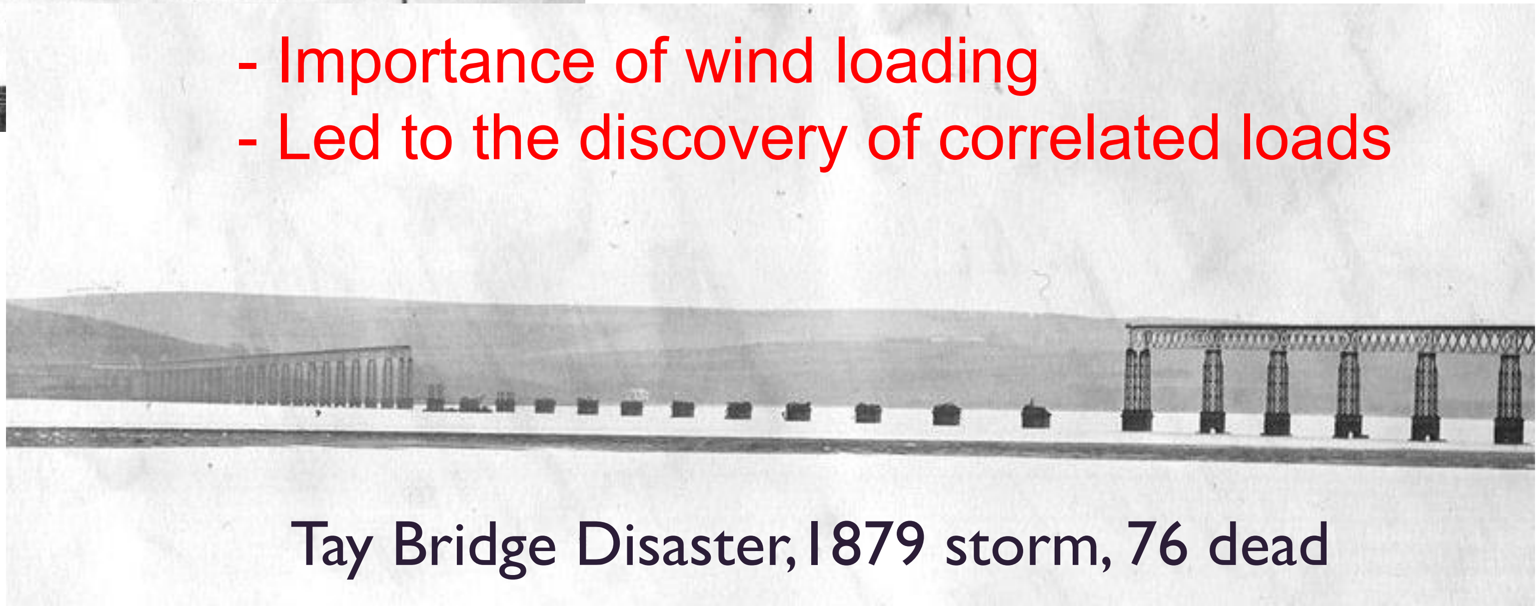


# Wind engineering

Tay Bridge



- Importance of wind loading
- Led to the discovery of correlated loads

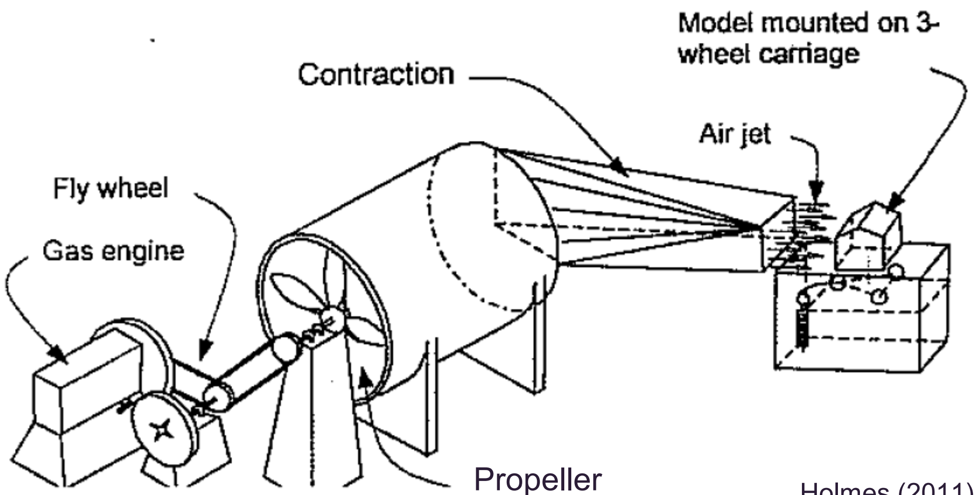


Tay Bridge Disaster, 1879 storm, 76 dead

# Wind Engineering

## The first wind tunnels

- **Kernot, 1893 in Melbourne, Australia**
  - Cubes, pyramids, cylinders, buildings.
- **Irminger, 1894 in Copenhagen, Denmark**




Holmes (2011)

Kernot's wind tunnel

# Wind tunnels

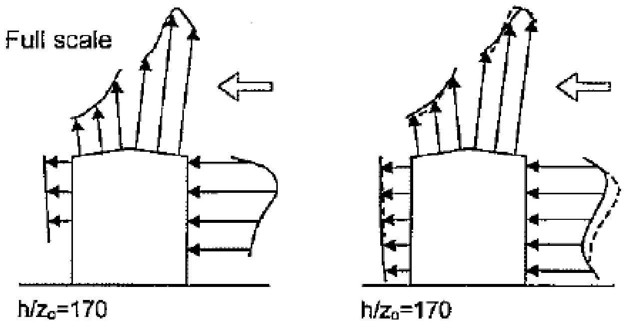
1930's and 40's at NPL full-scale pressure measurements on a shed were compared with turbulent and uniform wind tunnel tests.

- Pressure distribution varies greatly with turbulence and upwind characteristics.

1930's study of skyscrapers led to two significant conclusions.

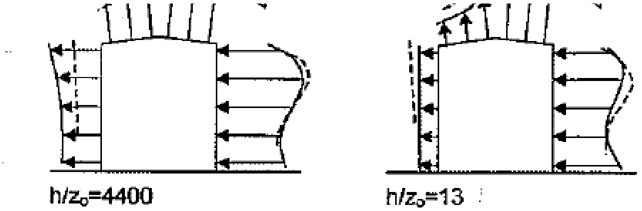
1. Wind speeds (and therefore pressures) increase rapidly with height,
2. Measured pressures are lower than in aeronautical wind tunnels.

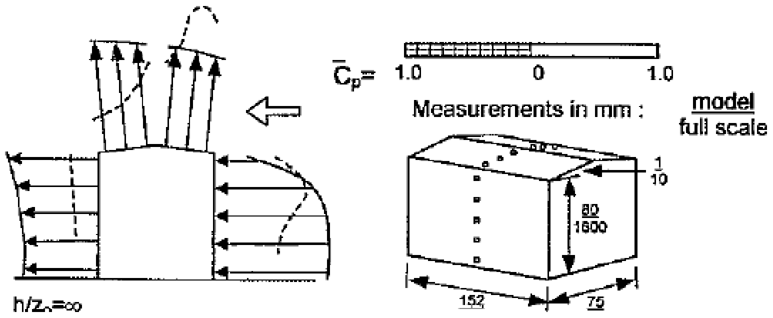
# Wind tunnels




Rough surface/high turbulence

Turbulence modifies how flow separates and reattaches to buildings




Jensen, 1950s, Denmark

Smooth surface/low turbulence

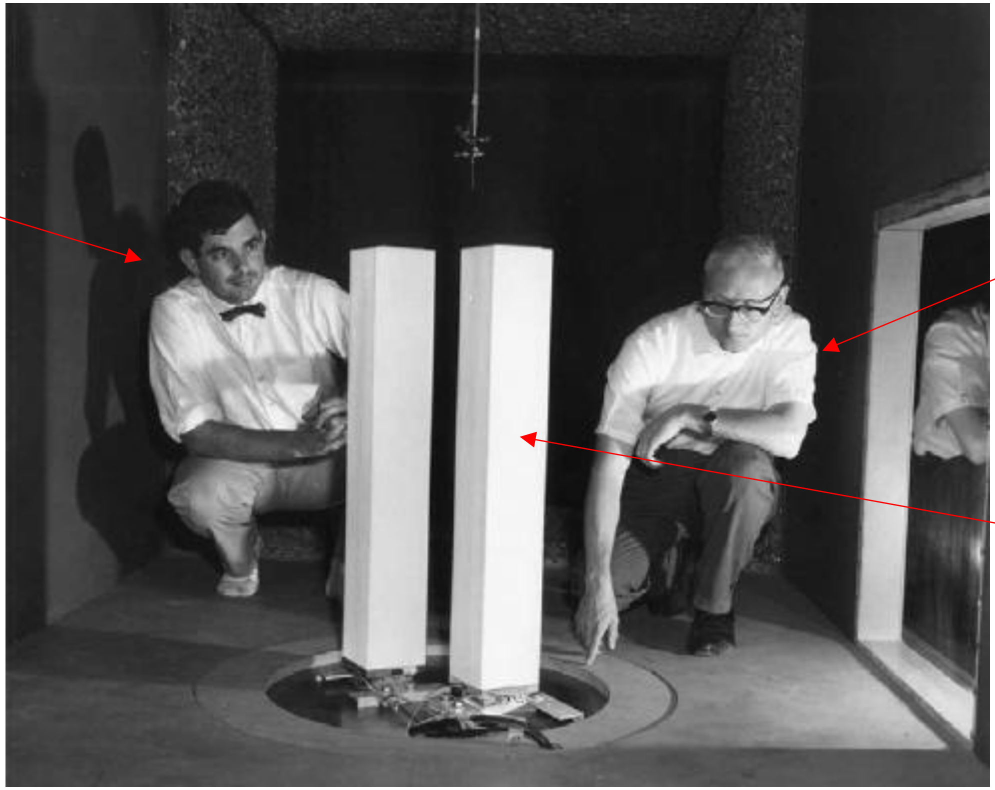
# Dynamic wind loading

Tacoma Narrows

Golden Gate Bridge

# Wind engineering post 1960

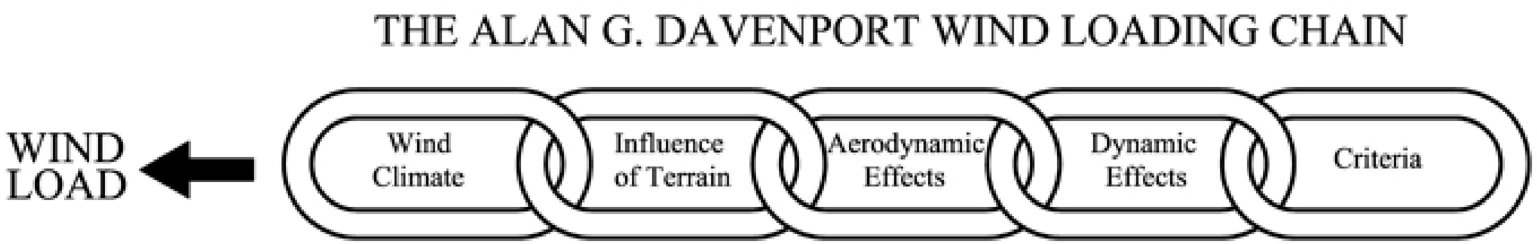
Alan Davenport



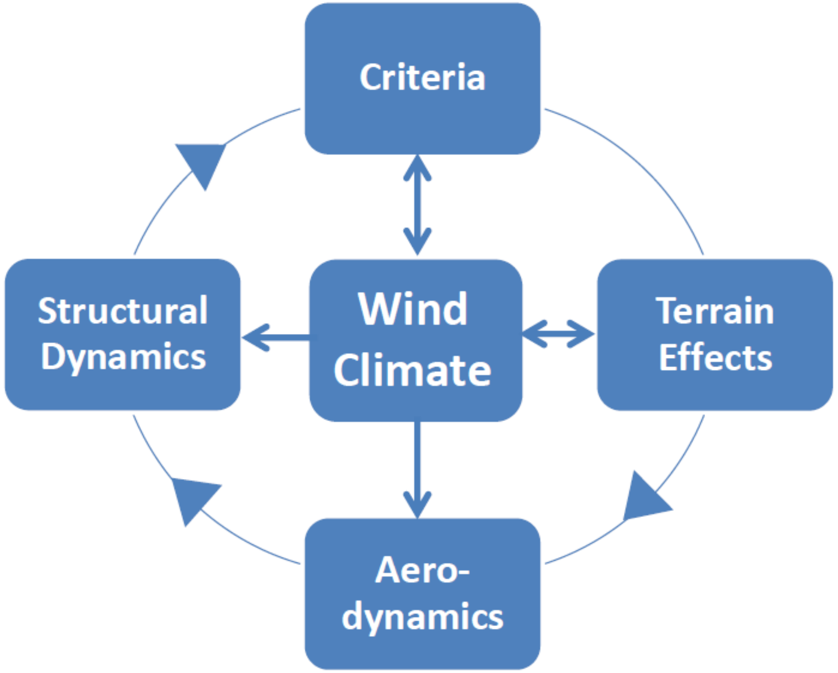
Jack Cermak

World Trade Center

# Bringing it all together




**Wind Loading Wheel**
Letchford and Lombardo (2015)

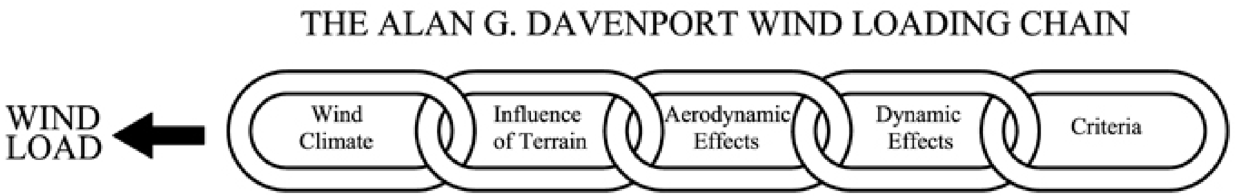


# Have we solved everything?

# A wind engineer's tools

## Wind loading of structures

- Wind tunnels
- Codes and Standards
- Online databases
- Computational fluid dynamics




## Measuring the wind

- Anemometers
- Doppler radars/Lidars
- Satellites

# *Meteorology of windstorms*

# Reading

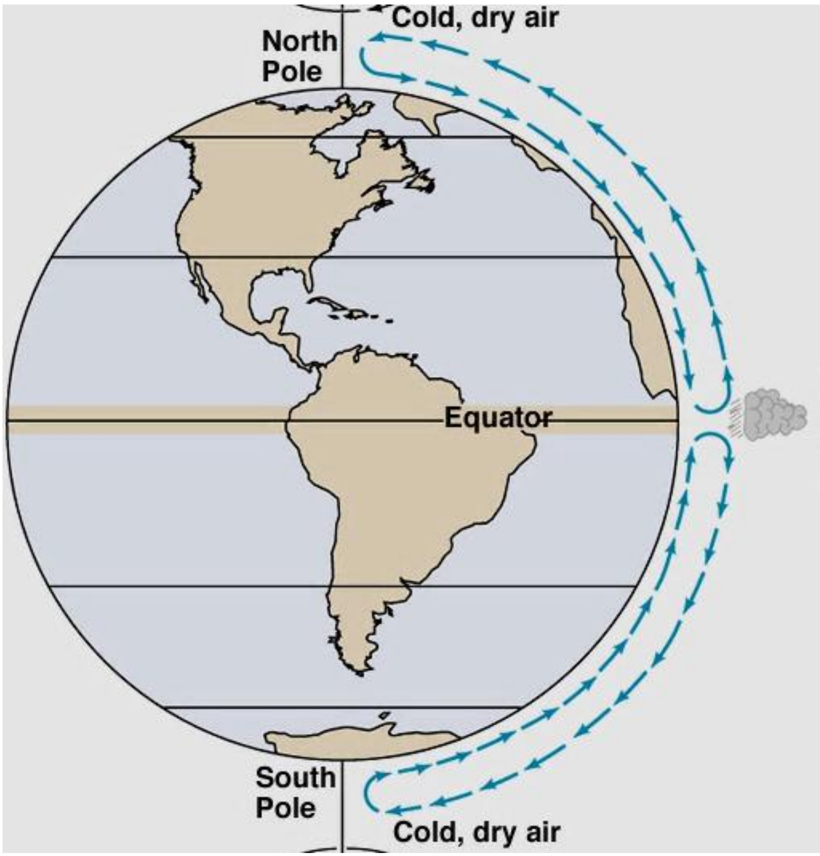
Holmes & Bekele – Chapter 1

Simiu & Yeo – Chapter 1

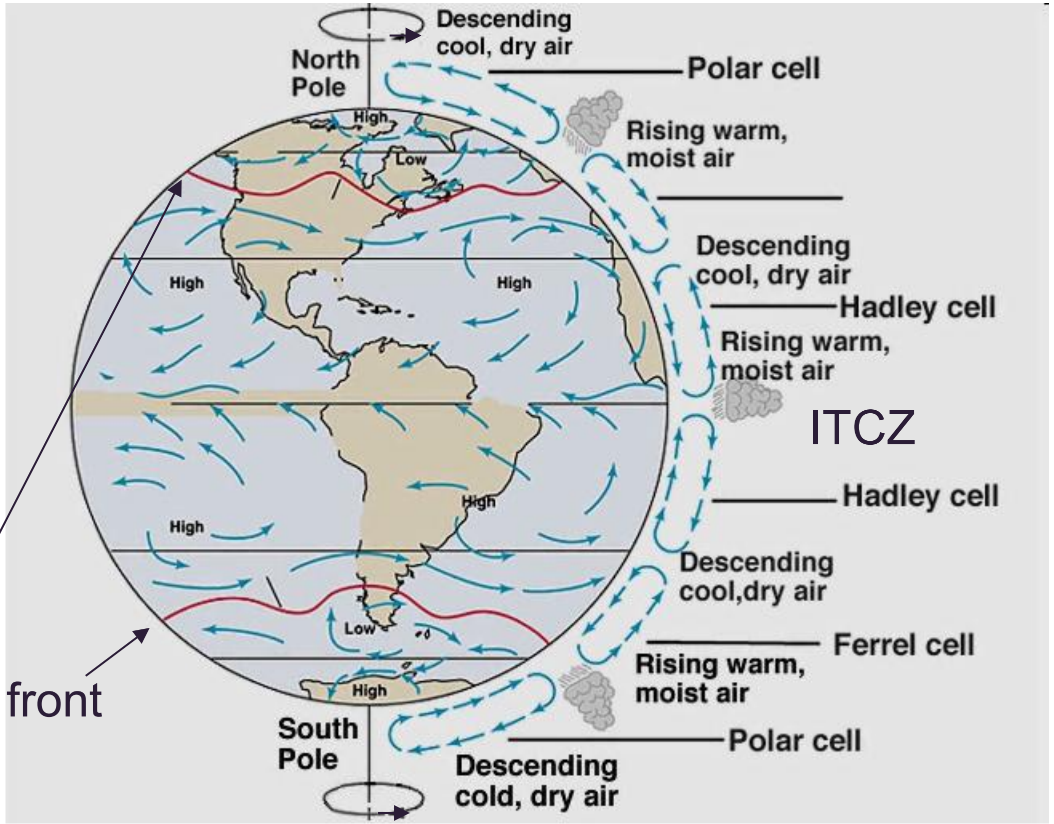
# Why does the wind blow?

# Basic meteorology

No rotation




Rotation




ITCZ

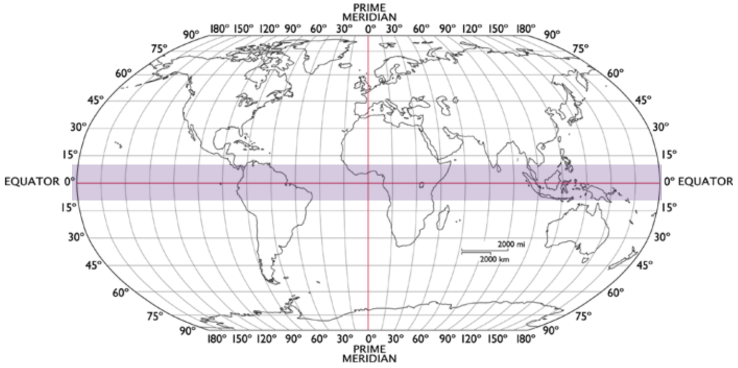
Polar front

http://ess.geology.ufl.edu/ess/Notes/AtmosphericCirculation/atmosphere.html

# Global (strong) winds

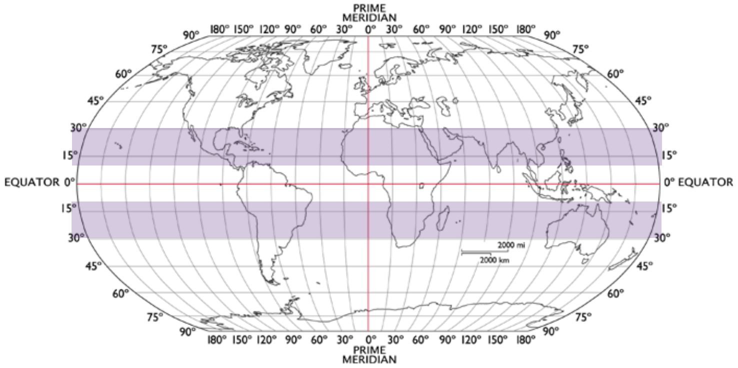
## Equatorial (+/-10°)

- Frequent thunderstorms and squalls
- Low wind speeds
- Little tendency for cyclonic systems to develop (Coriolis)



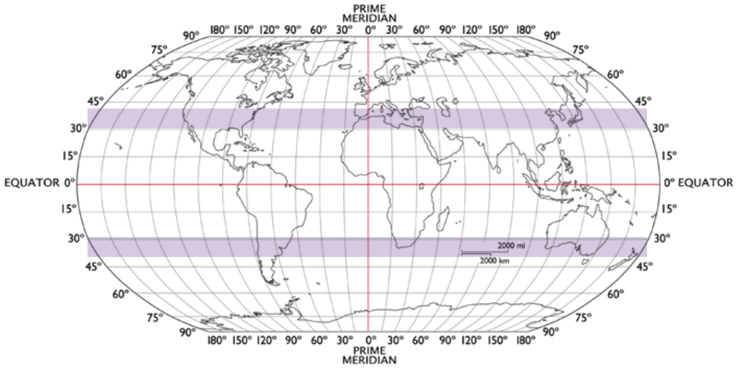

## Tropical region (10°- 30°)

- Seasonal monsoon rains – not strong winds
- Sea surface temperatures (and Coriolis forces) are high enough for Tropical Cyclones




## Sub-tropical region (30°- 40°)

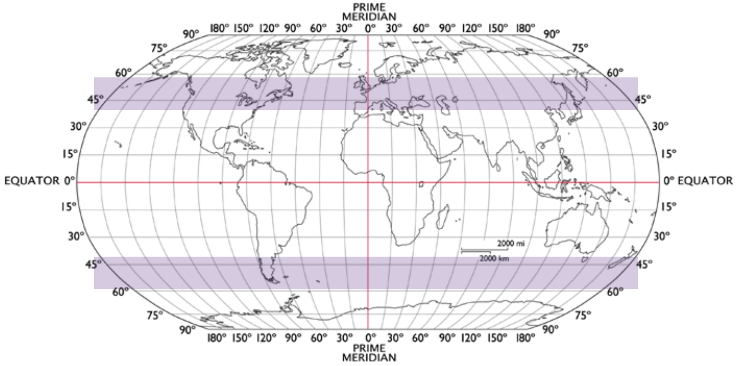
- Central Australia, South Africa, Southern US, Russia, central south America
- High pressure zone
- Thunderstorms are common (solar heating)



# Global (strong) winds

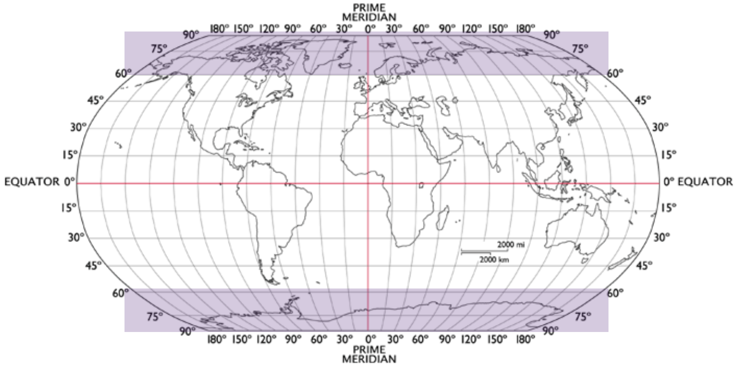
## Temperate region (40°- 60°)

- Southern South America and Australia, New Zealand, Europe, North America, and Asia
- Unstable frontal systems develop with travelling lows
- Thunderstorms are common and extratropical cyclones can occur

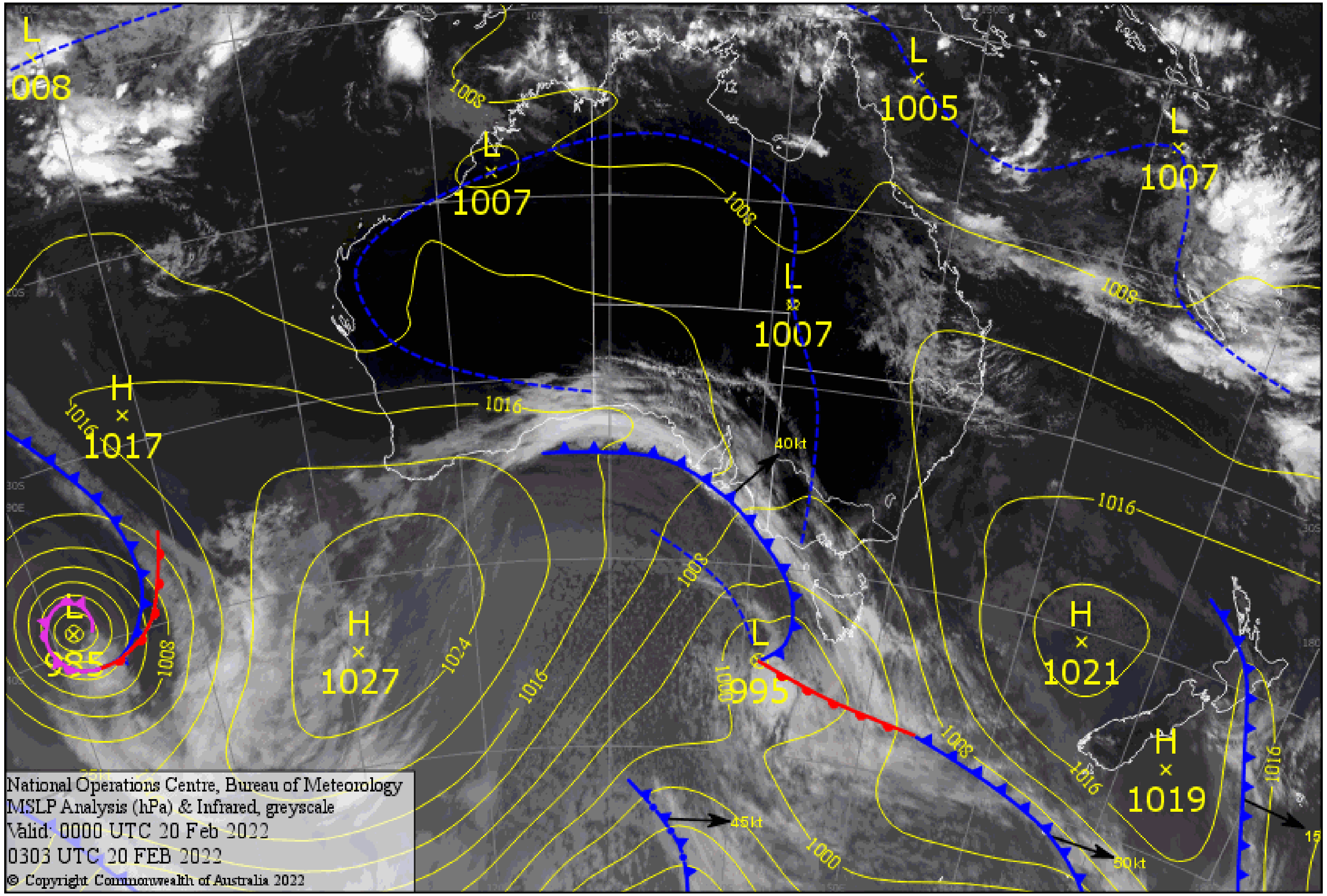



## Polar region(60°- 90°)

- Antarctic, Artic, Scandinavia, Russia, Canada
- Dynamic pressure increased due to higher air density
- Strong steady katabatic winds occur on steep slopes



# Localised weather




http://www.bom.gov.au/

# Windstorms of interest to wind engineers

- Synoptic gales (extratropical cyclones)
- Frontal systems
- Tropical cyclones
- Thunderstorms
- Tornadoes
- Other winds

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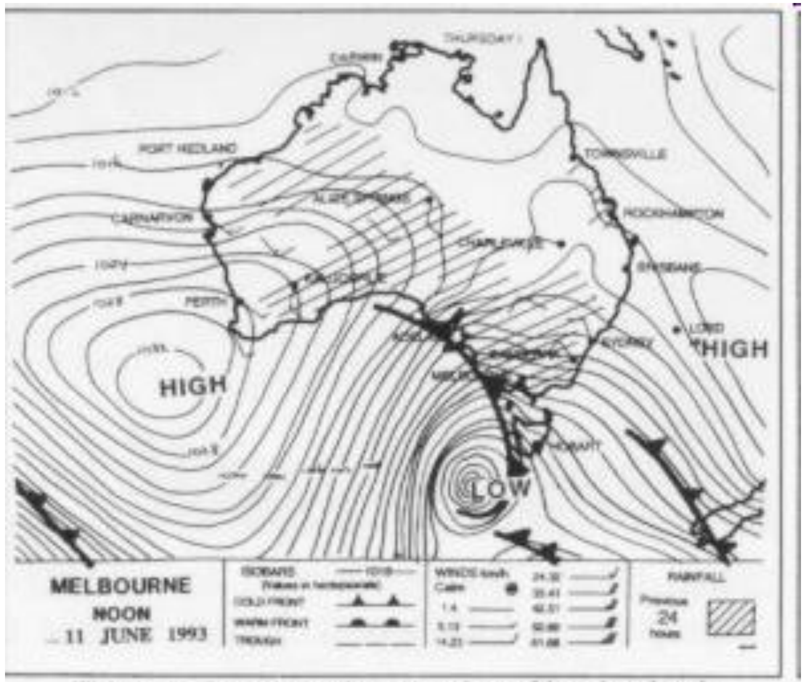
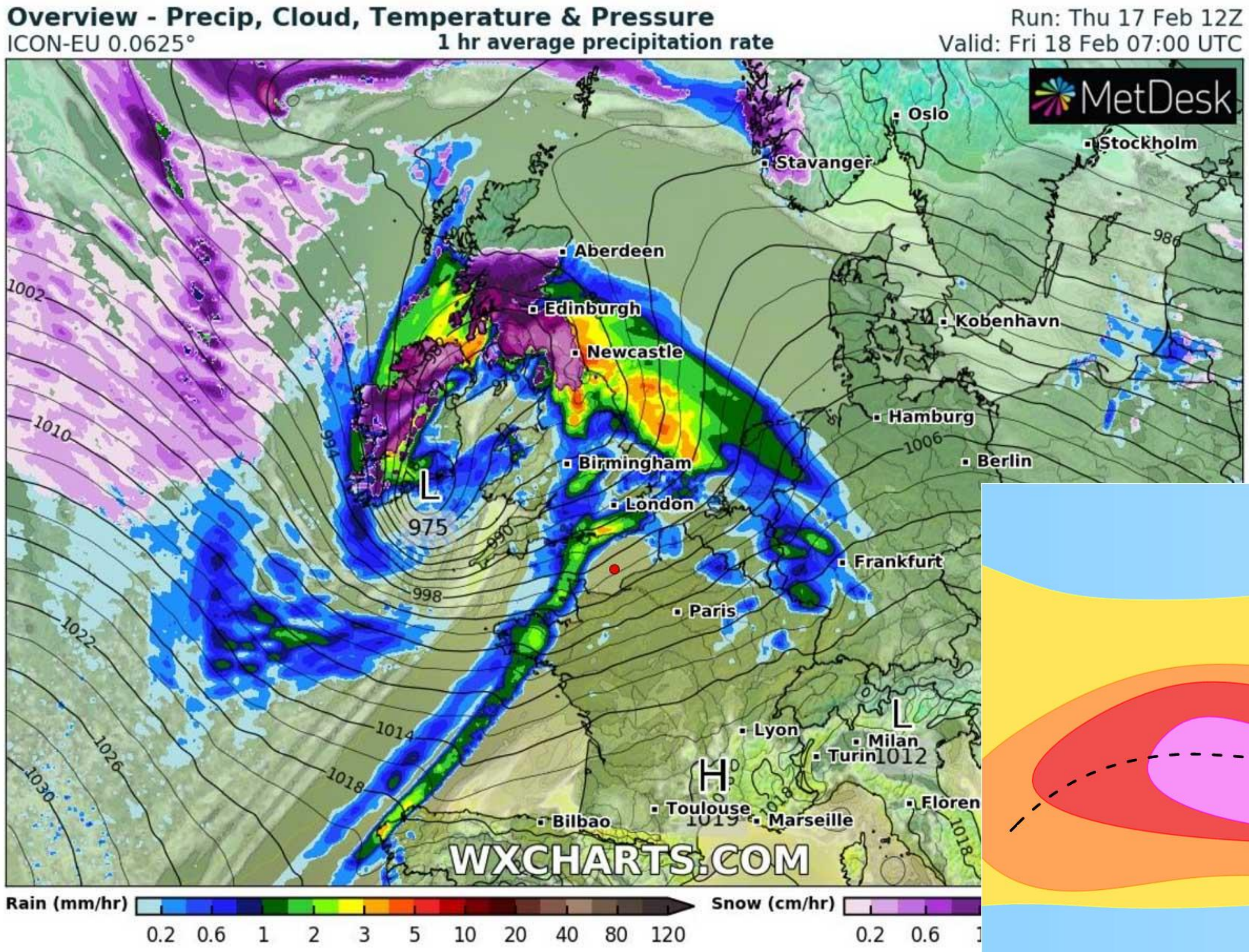
# Synoptic gales

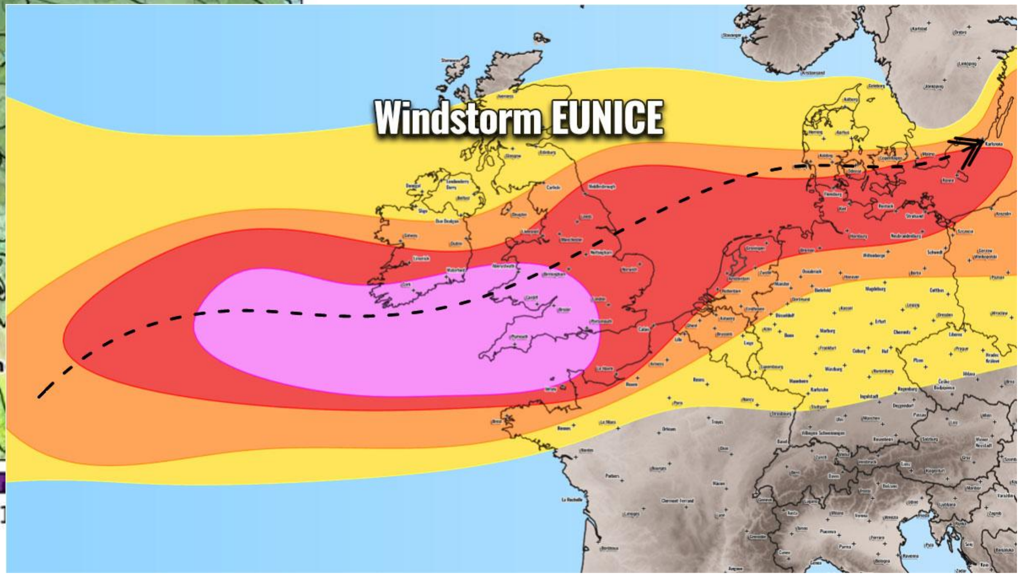



This weather map shows a day of land gales in southern Victoria. Close spacing of the lines (Isobars) around the LOW indicates a large pressure difference over a small distance

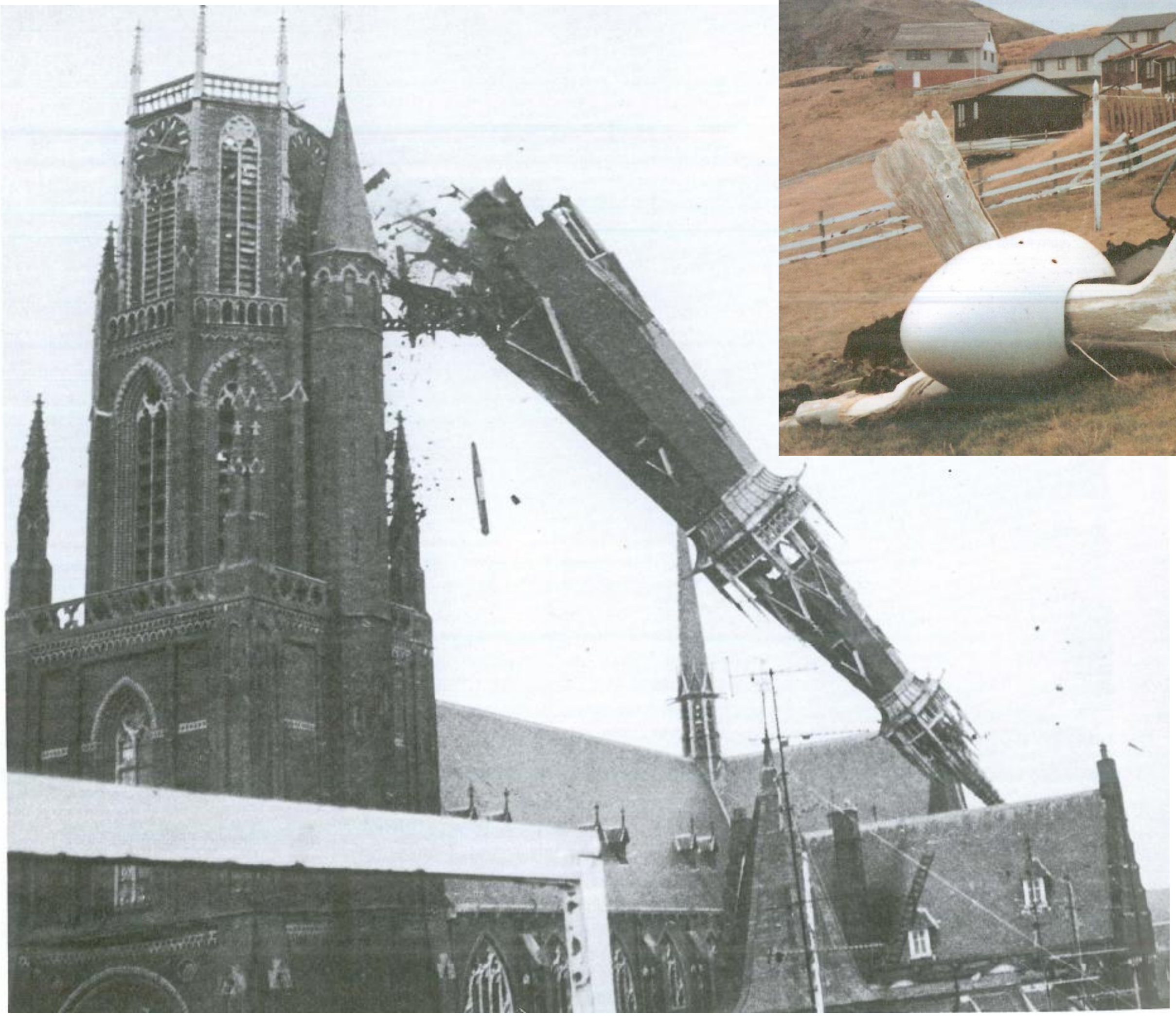
# Synoptic gales




Windstorm Eunice

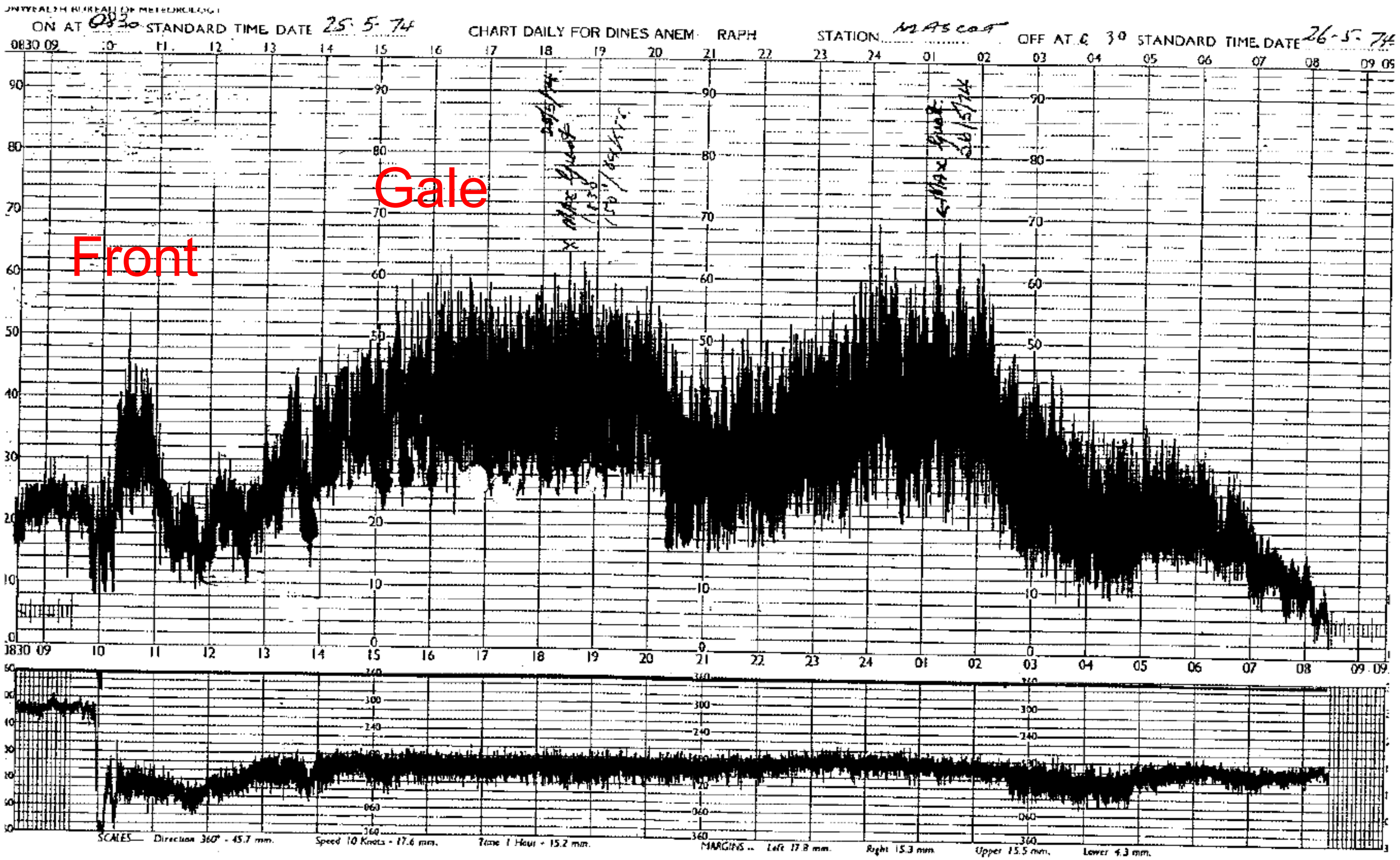


# Synoptic gales - damage



Munich Re.

# Synoptic gales - structure

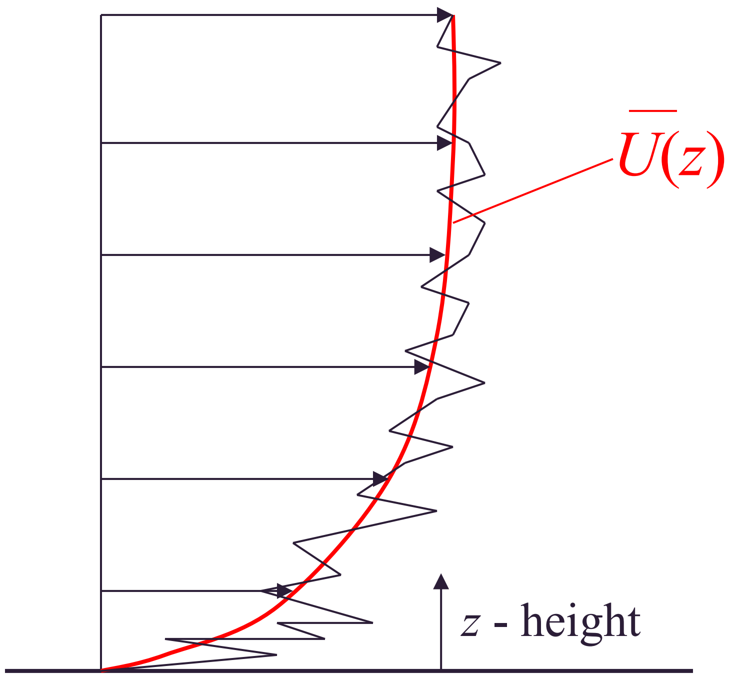


# Wind - structure

- Wind speed and direction fluctuate.
- Wind speed increases with height.
- Turbulence decreases with height.
- Large range of frequencies in the wind.
- Correlation between locations exists.

- **All depend on type of storm!**

# Synoptic gales - structure




Prandtl's Logarithmic Law (<200-300 m)

$$\overline{U}(z) = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right)$$

Von Karman constant

Aerodynamic roughness

$$u_* = \sqrt{\frac{\tau_0}{\rho}}$$

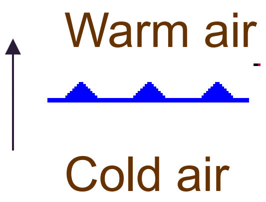
Surface shear stress

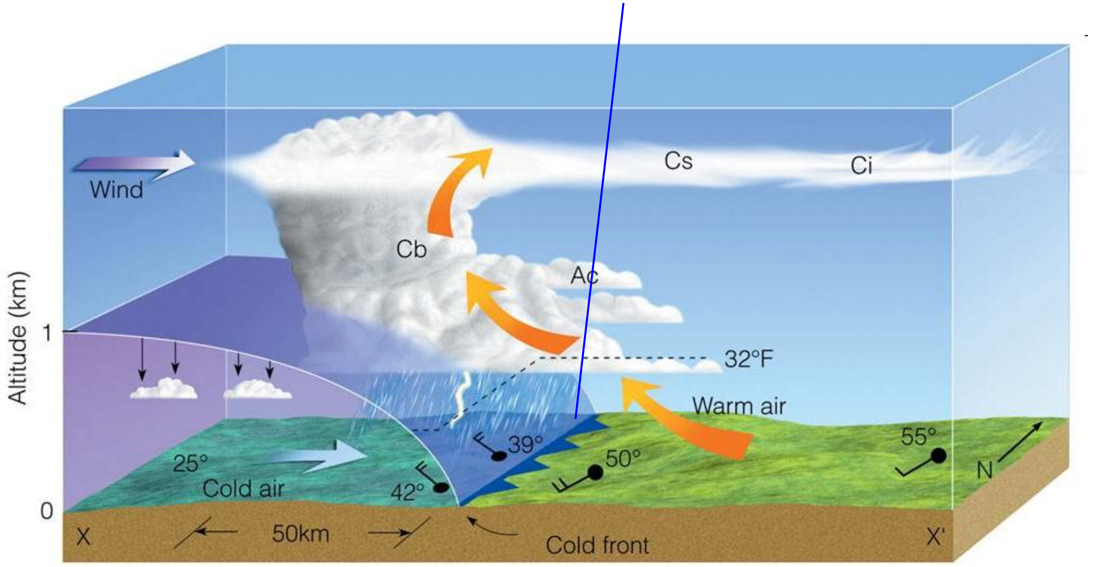
Friction velocity

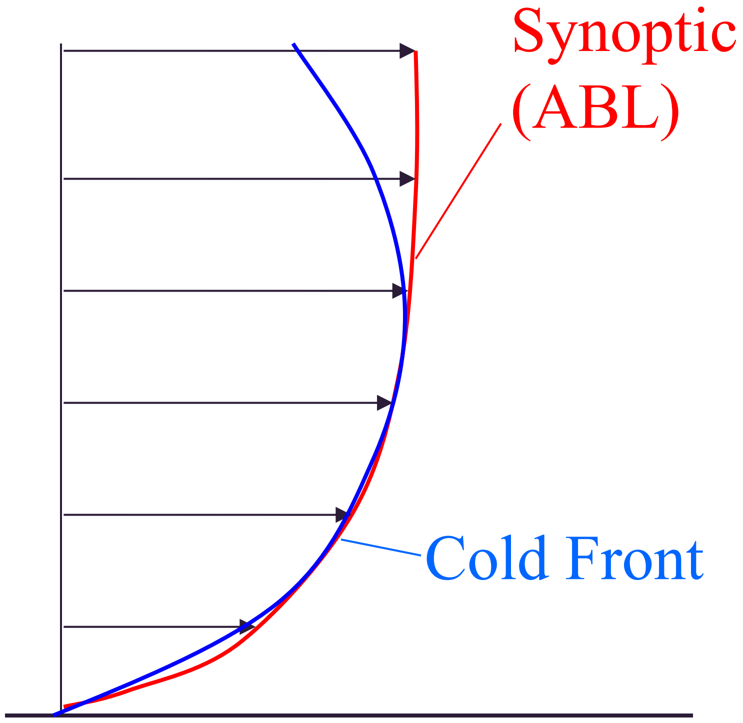
Fluid density

# Frontal systems

## Cold front



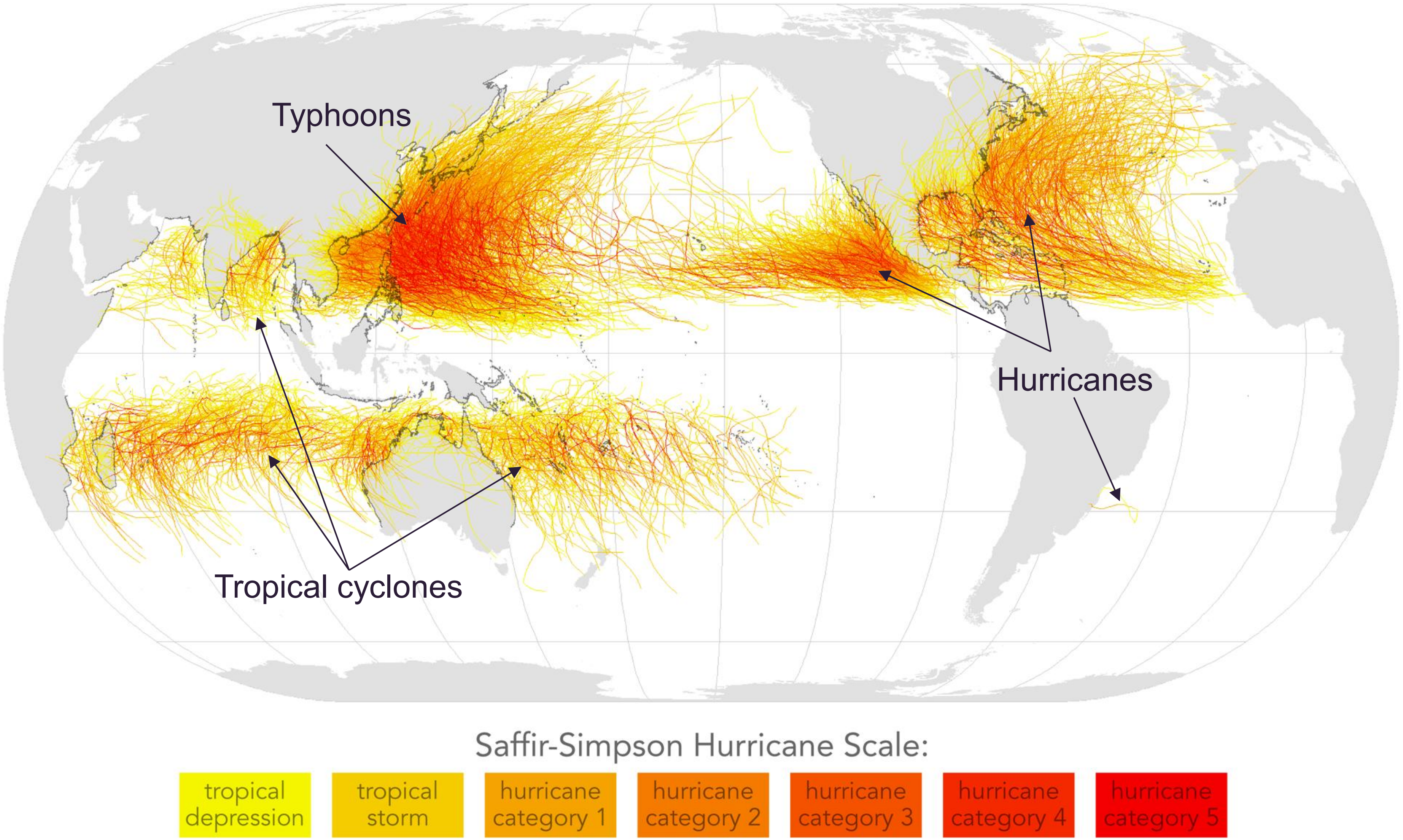




http://apollo.lsc.vsc.edu/classes/met130/notes/chapter11/

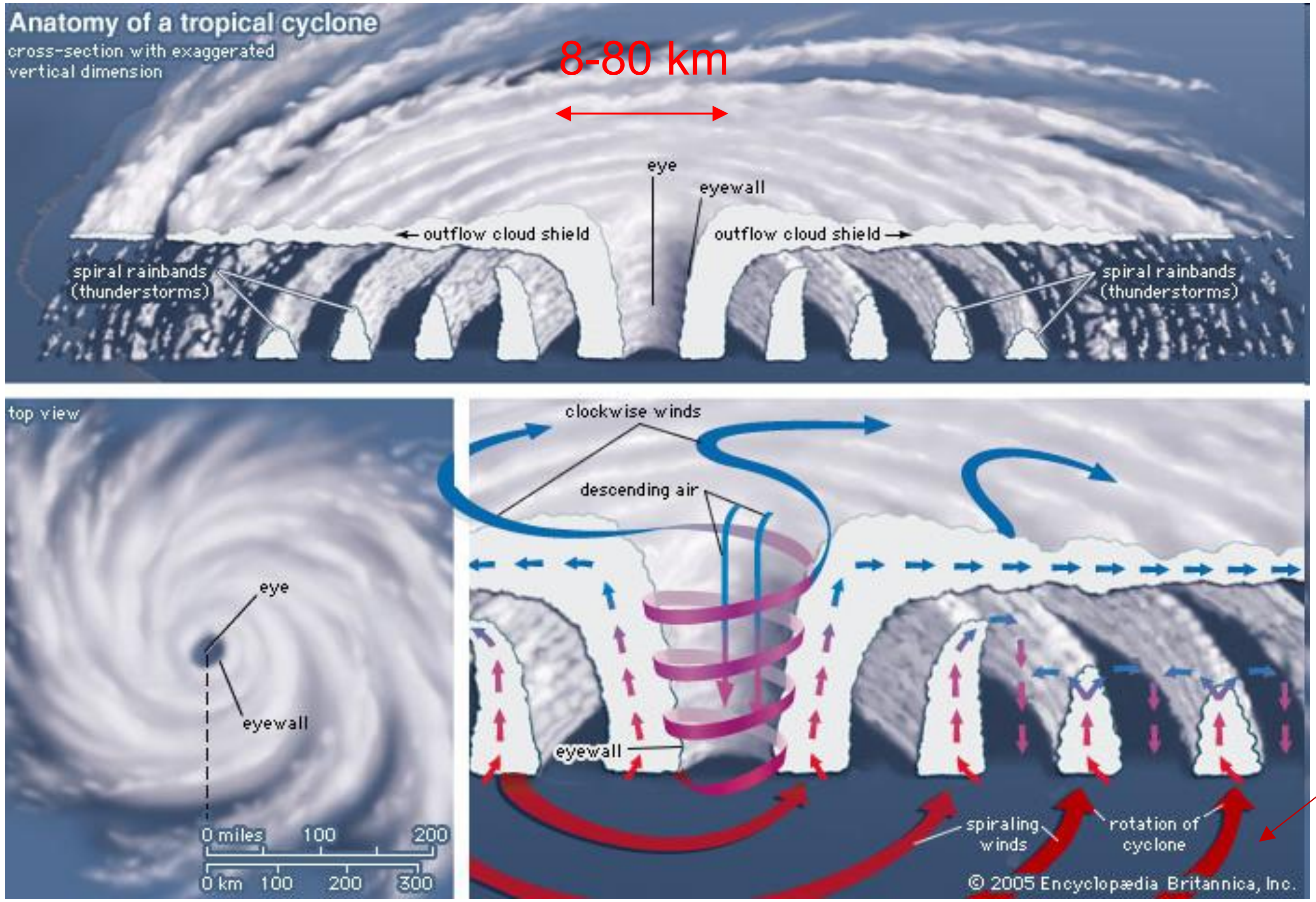
# Tropical Cyclones, 1945–2006






http://upload.wikimedia.org/wikipedia/en/0/0f/Tropical_cyclones_1945_2006.png

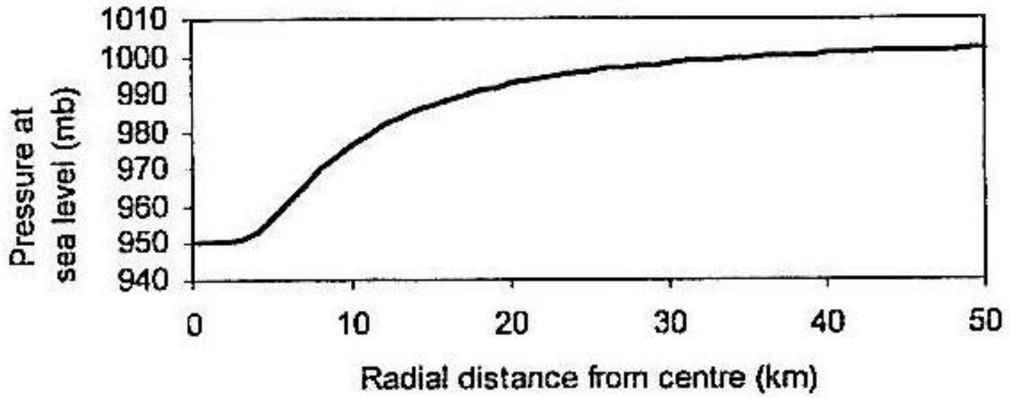
# Tropical cyclones - structure

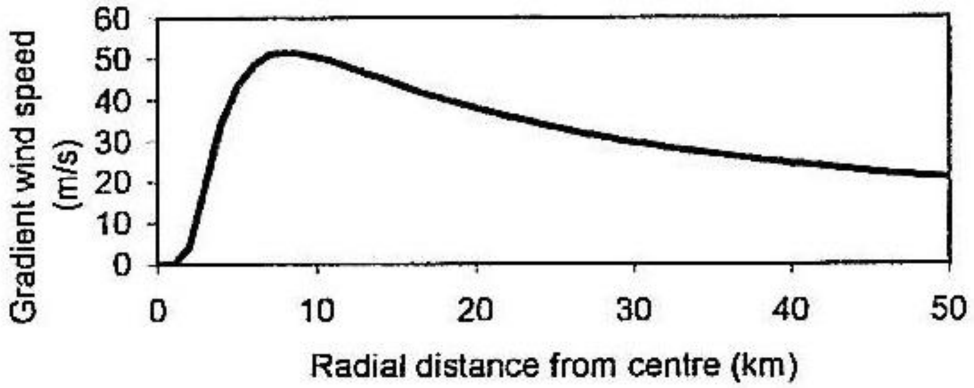



*Southern hemisphere rotation is opposite*

http://media-2.web.britannica.com/eb-media/44/76744-004-AF8CF4A2.jpg

# Tropical cyclones - structure



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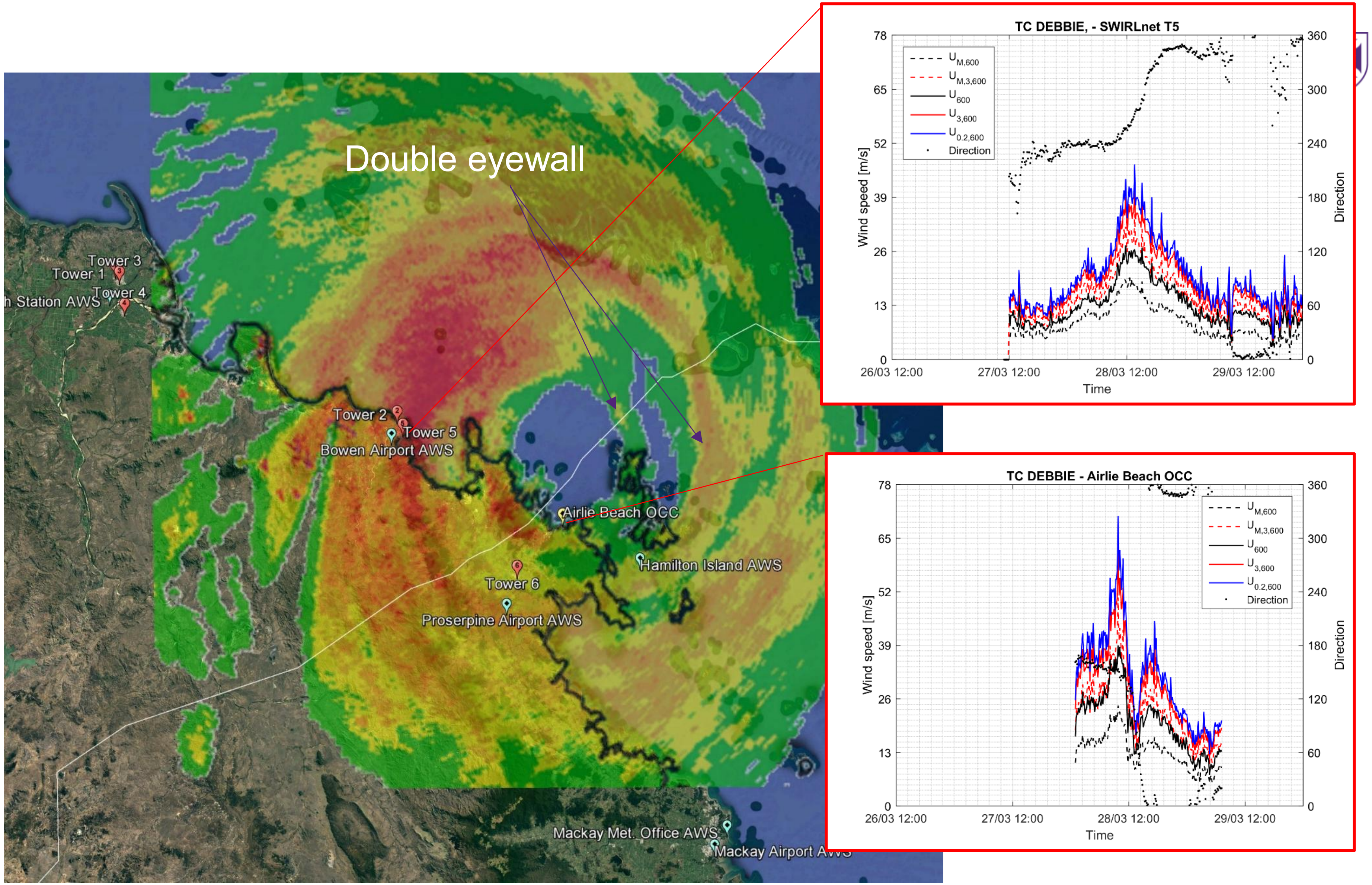
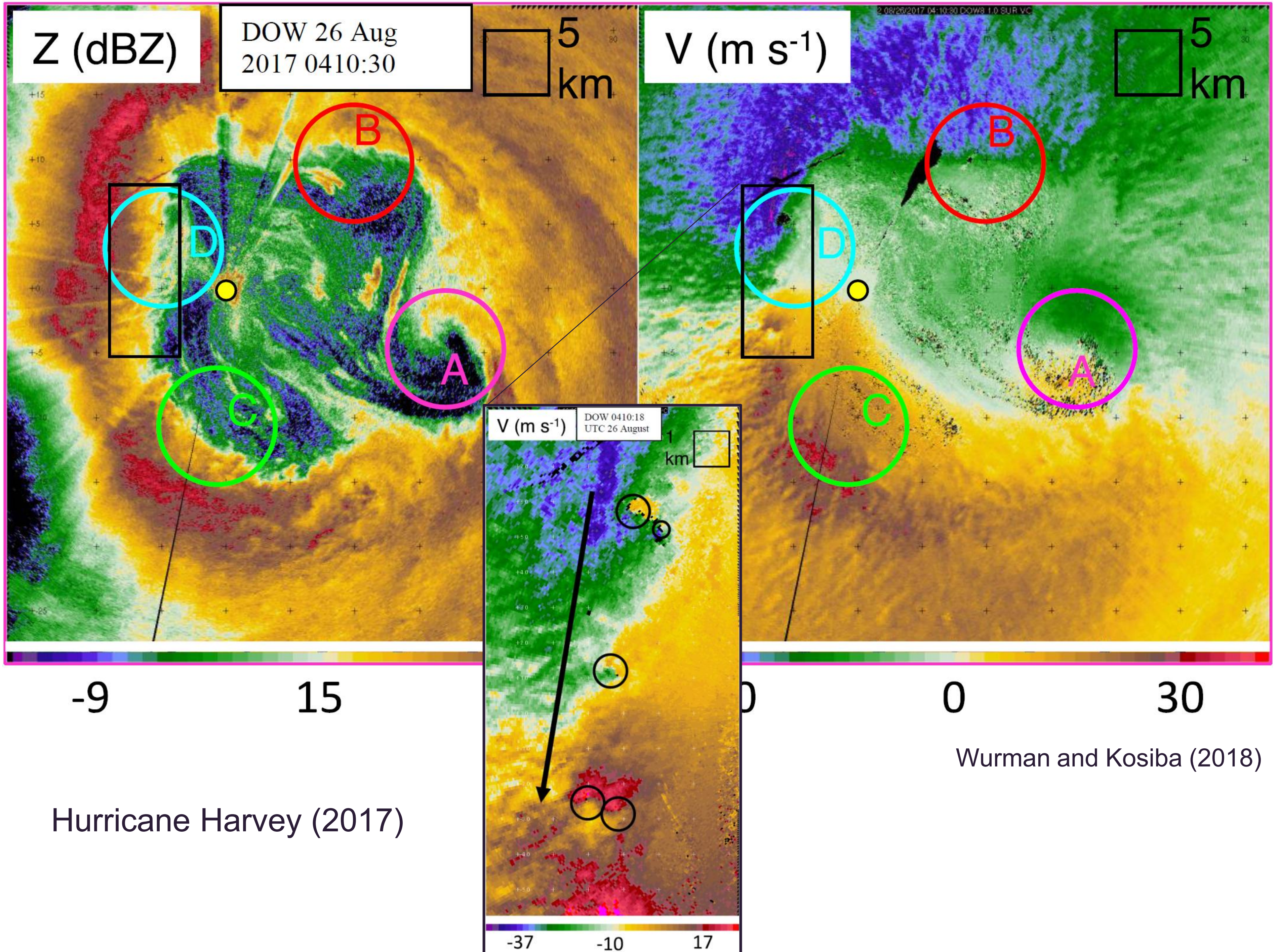



Cyclone Debbie (2017)

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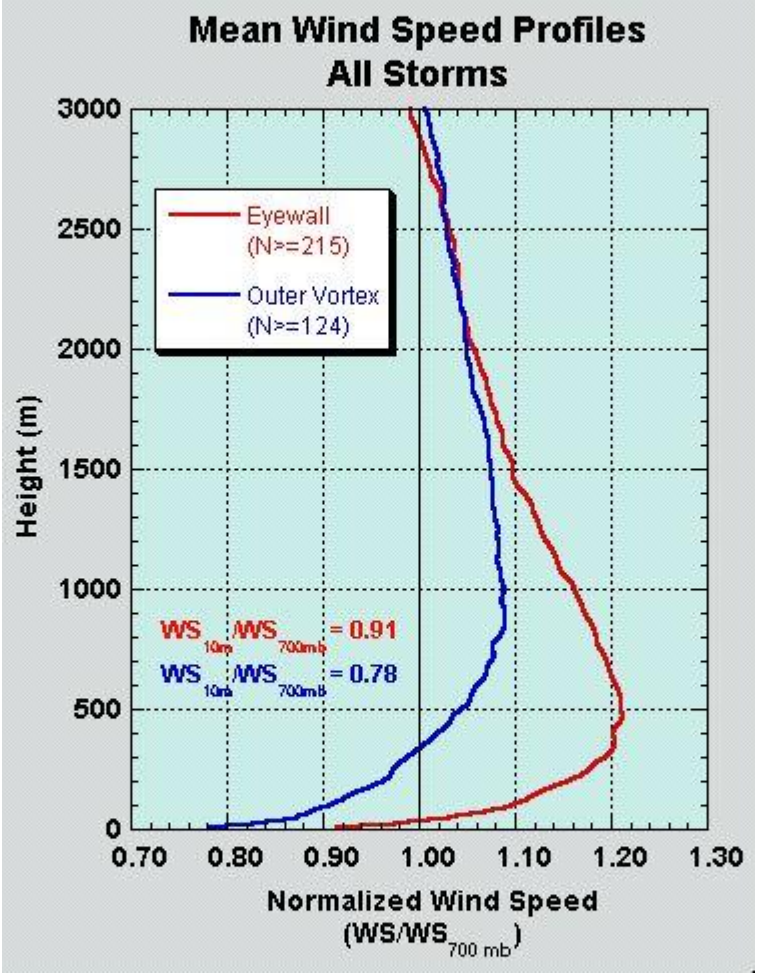
Wurman and Kosiba (2018)

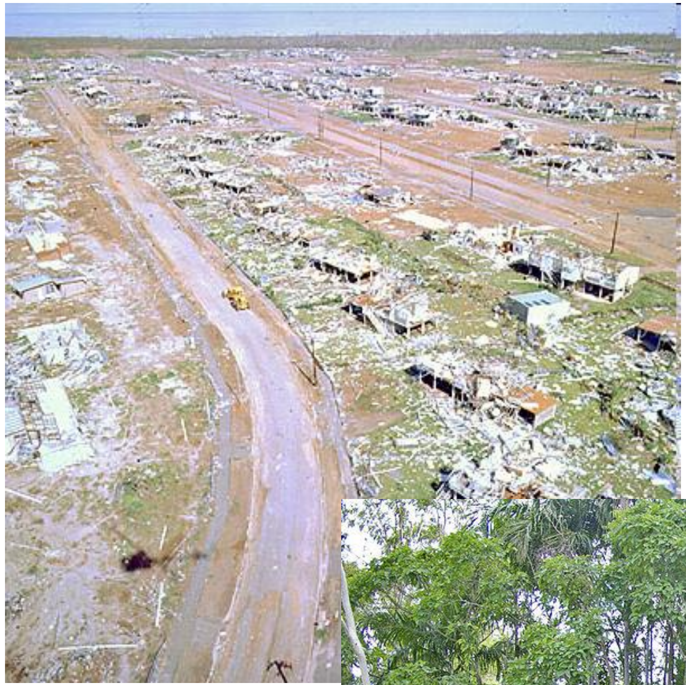
Hurricane Harvey (2017)

# Tropical cyclone - structure




http://www.nhc.noaa.gov/gifs/JLF_Fig1.jpg

# Tropical cyclones - damage



Cyclone Tracy (1974)

http://en.wikipedia.org/wiki/Cyclone_Tracy

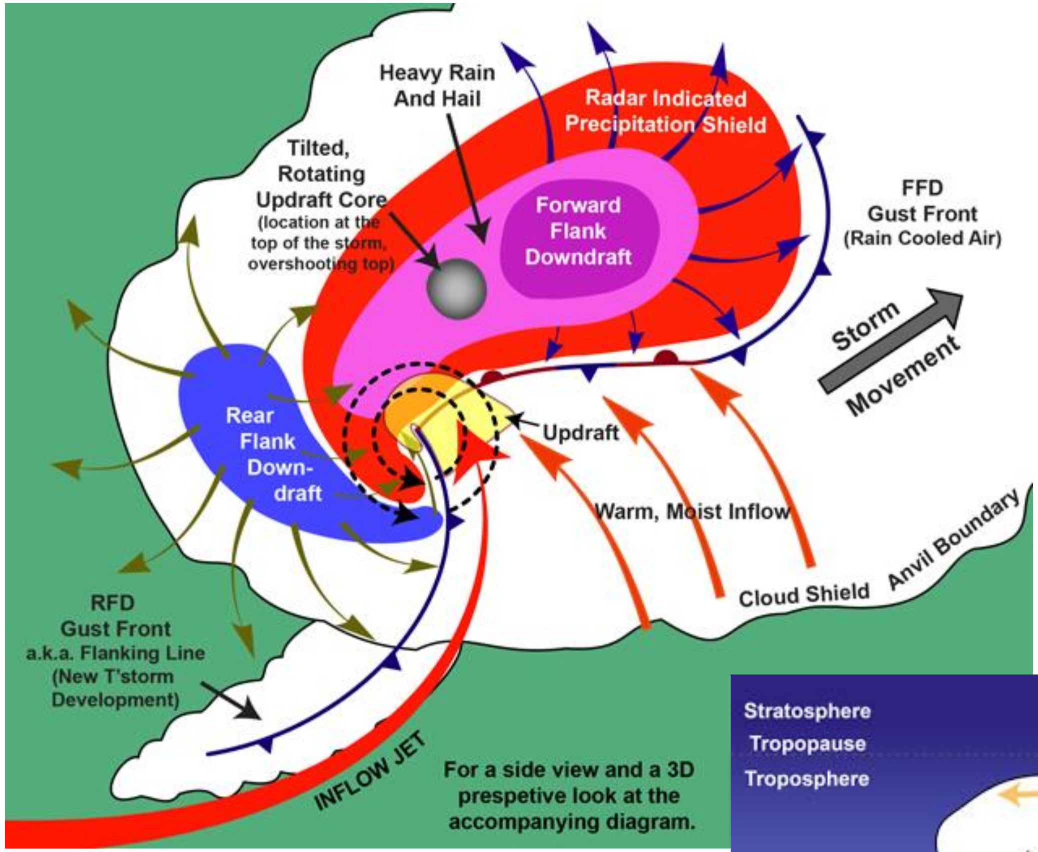
# Cyclone Yasi (2011)



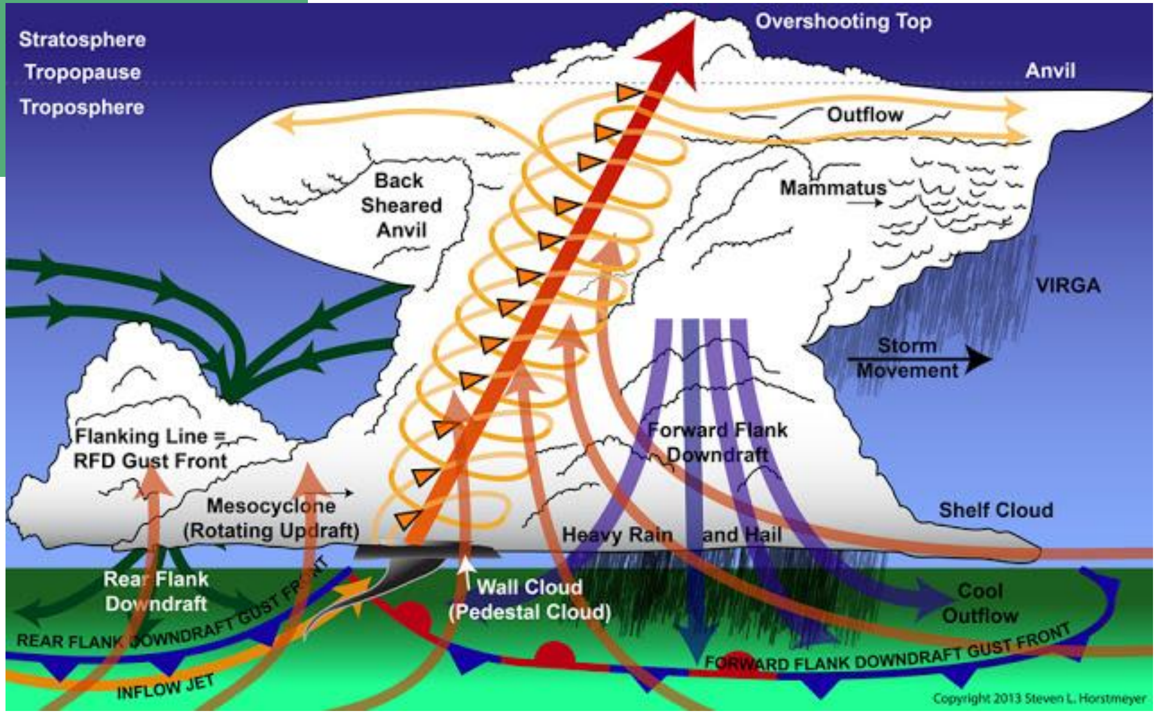
Tracy

# Thunderstorms

- Relatively small scale – hundreds of metres to kms.
- Short time duration – minutes to hours.
- Rapidly changing velocity and direction.
- Generally need.
  - Lifting mechanism – surface heating, front, sea-breeze, topography.
  - Shear
  - Moist air.
- Downbursts (outflow) and tornadoes (inflow).

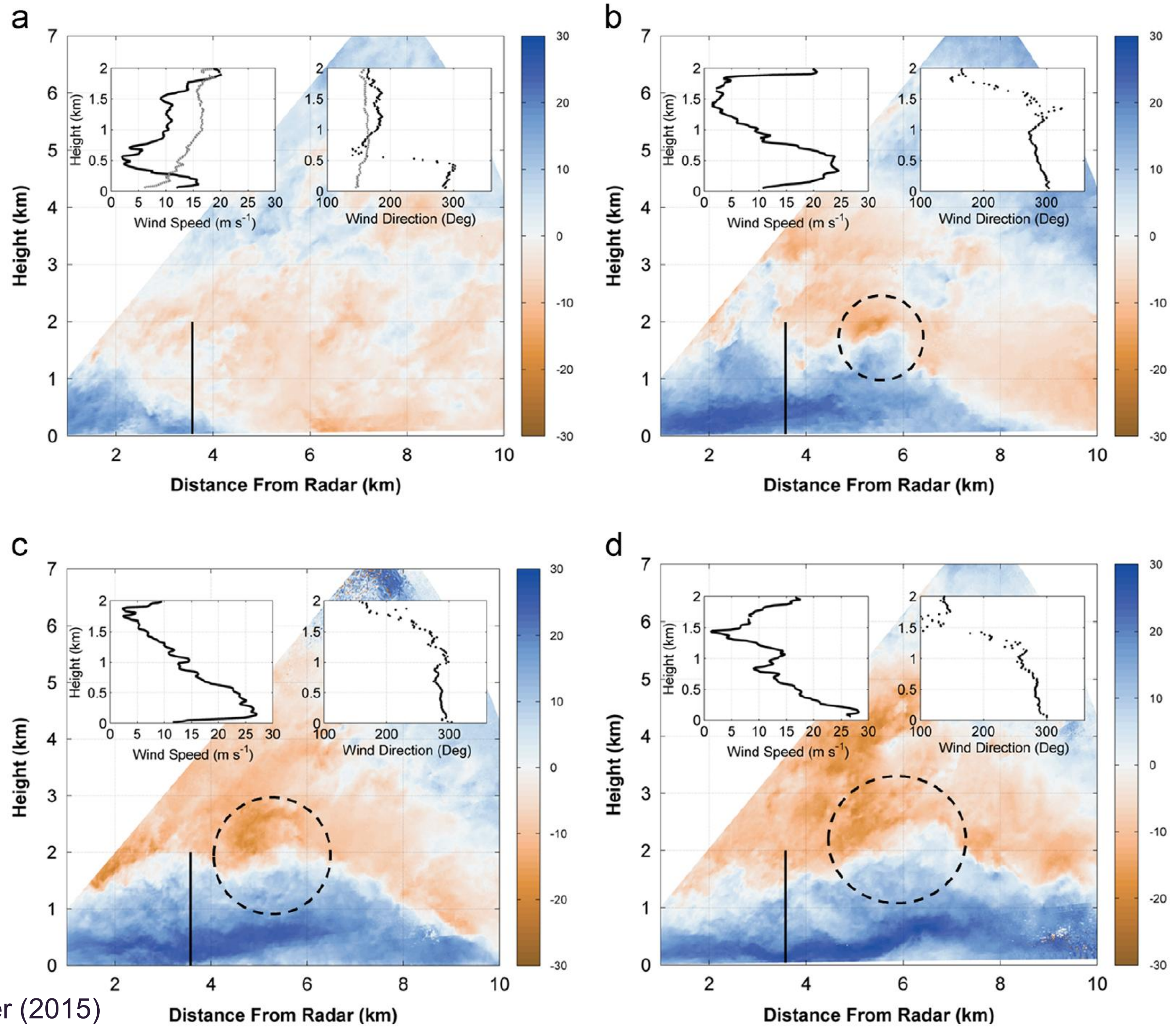



Structure of a supercell thunderstorm (Northern Hemisphere)



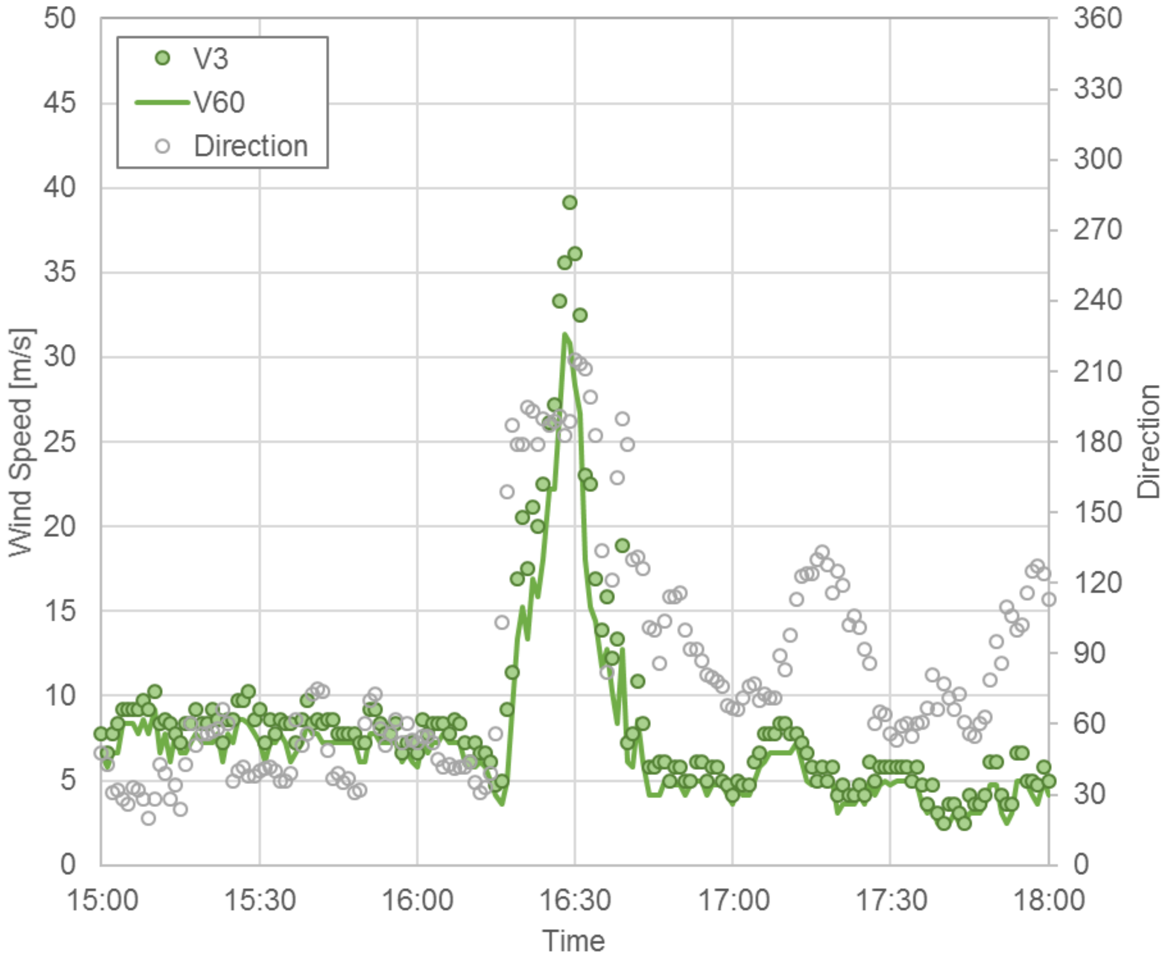
# Downburst (outflow)

# Downbursts - structure



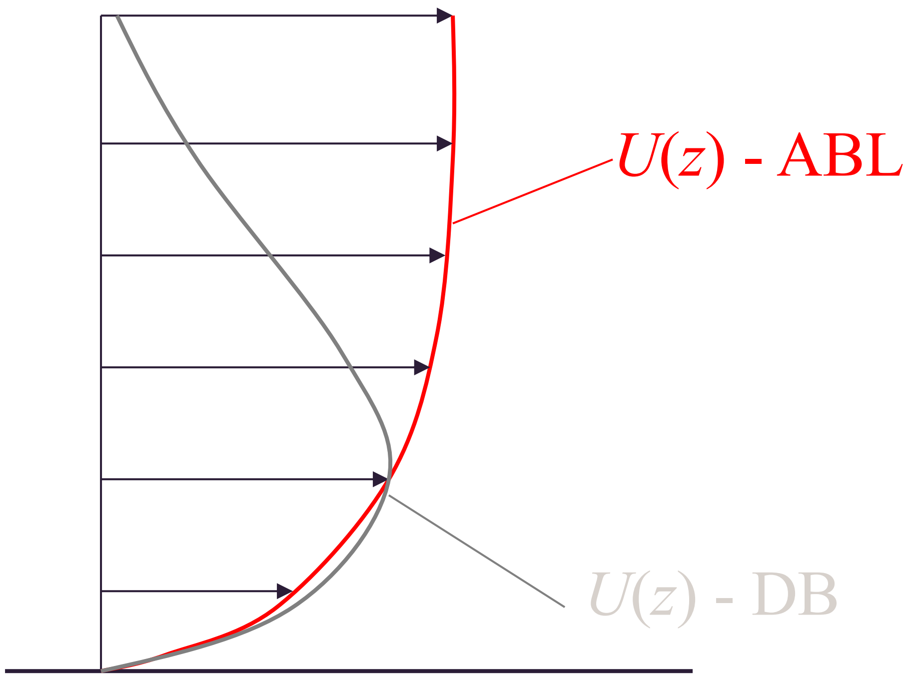

Gunter and Schroeder (2015)

# Downbursts - structure




Brisbane Hailstorm, 2014

# Downbursts - structure



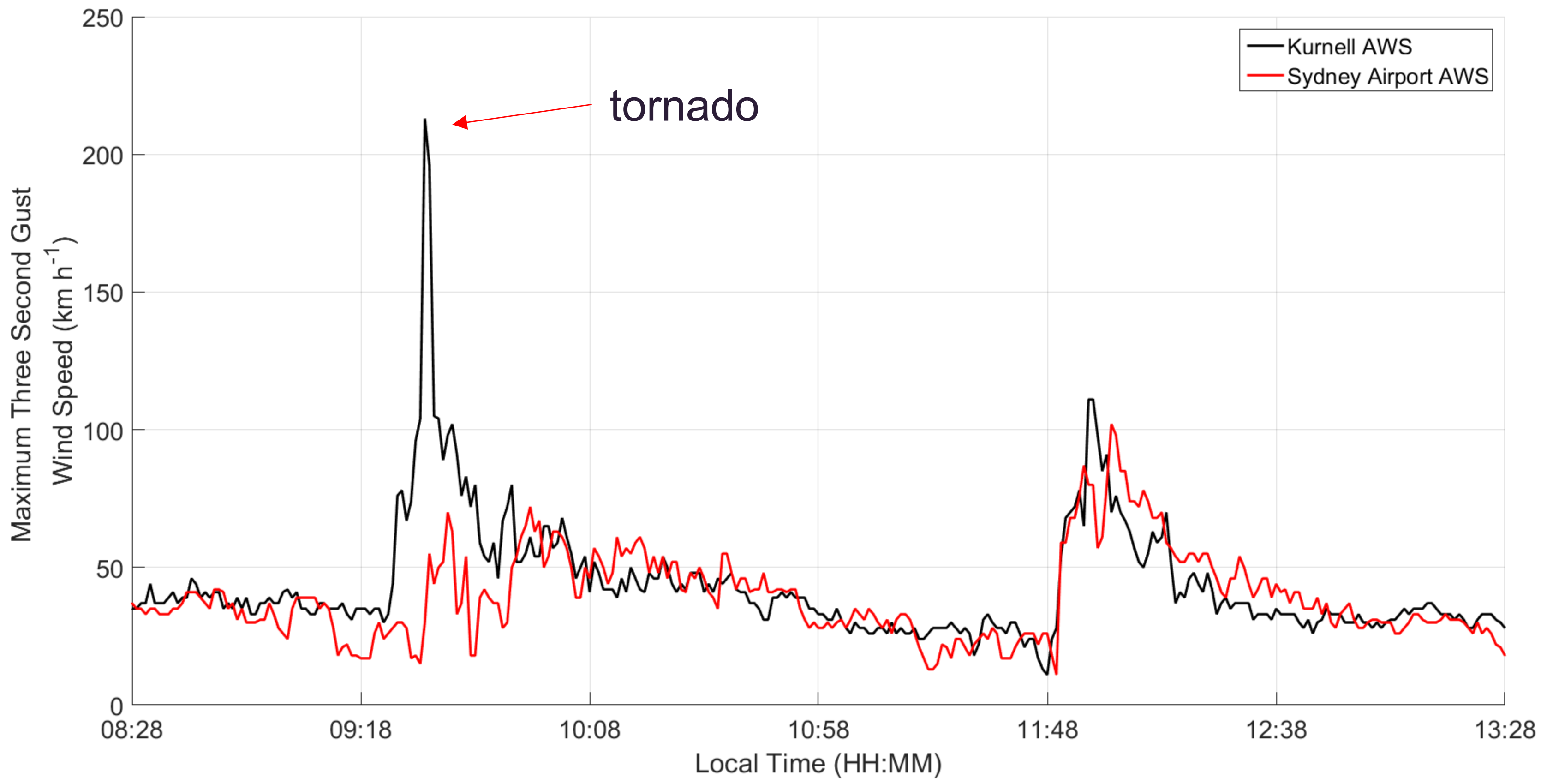

- Velocity profile can peak below 150 m, then drop off rapidly.
- Little full-scale evidence so is hard to validate experiments.
- Rapid change in velocity means that mean velocities are less meaningful than in synoptic scale storms.

# Downbursts - damage

# Tornadoes

Reed Timmer https://www.youtube.com/watch?v=T7auUE3PWXU

# Tornadoes




Kurnell tornado, 2015

# Tornadoes - structure

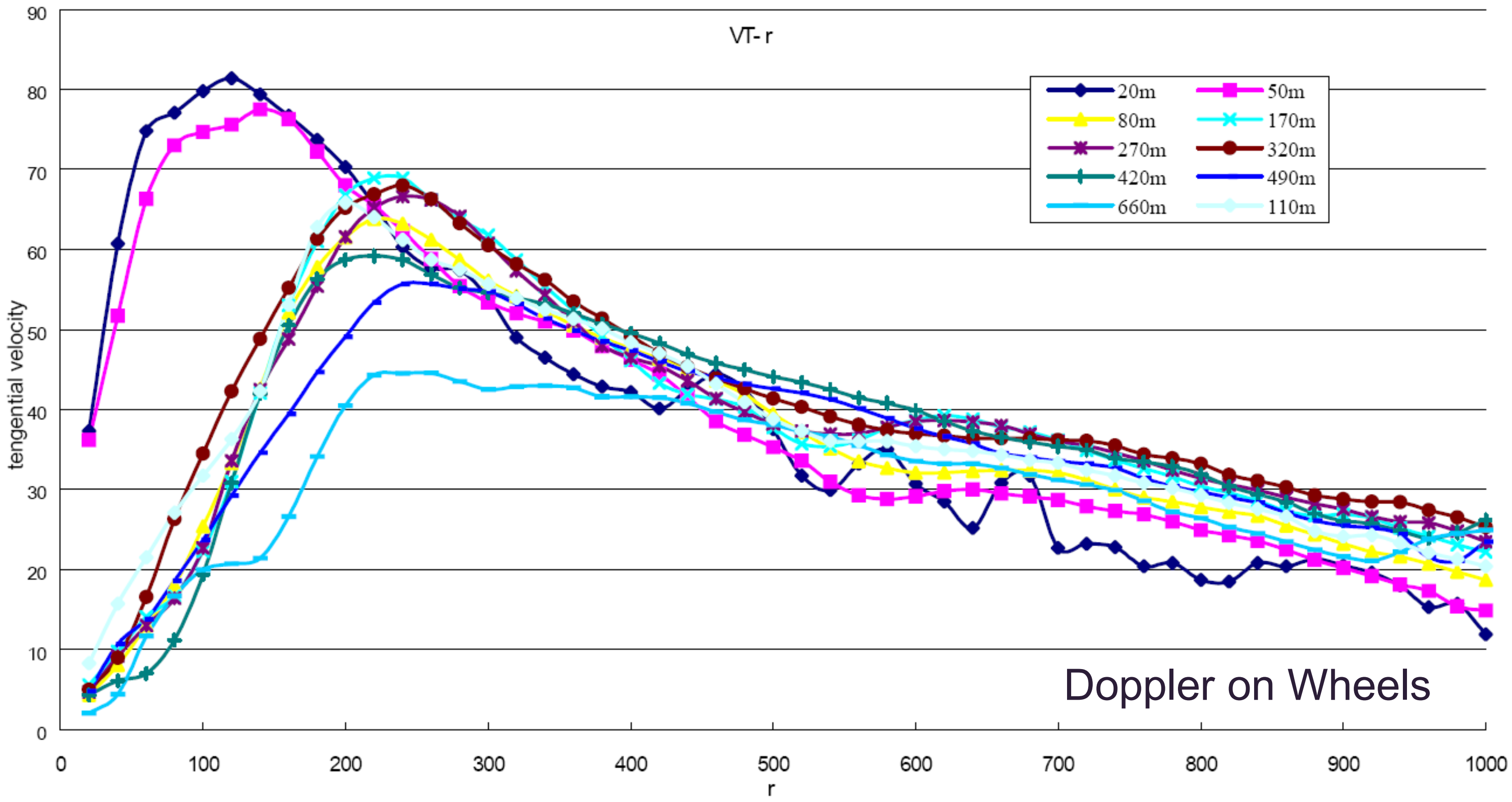



Figure 3: Radar-observed tangential velocity profiles in m/s (averaged azimuthally) as a function of radial distance (meters). Different colored curves show profiles at different elevations (m) above ground.

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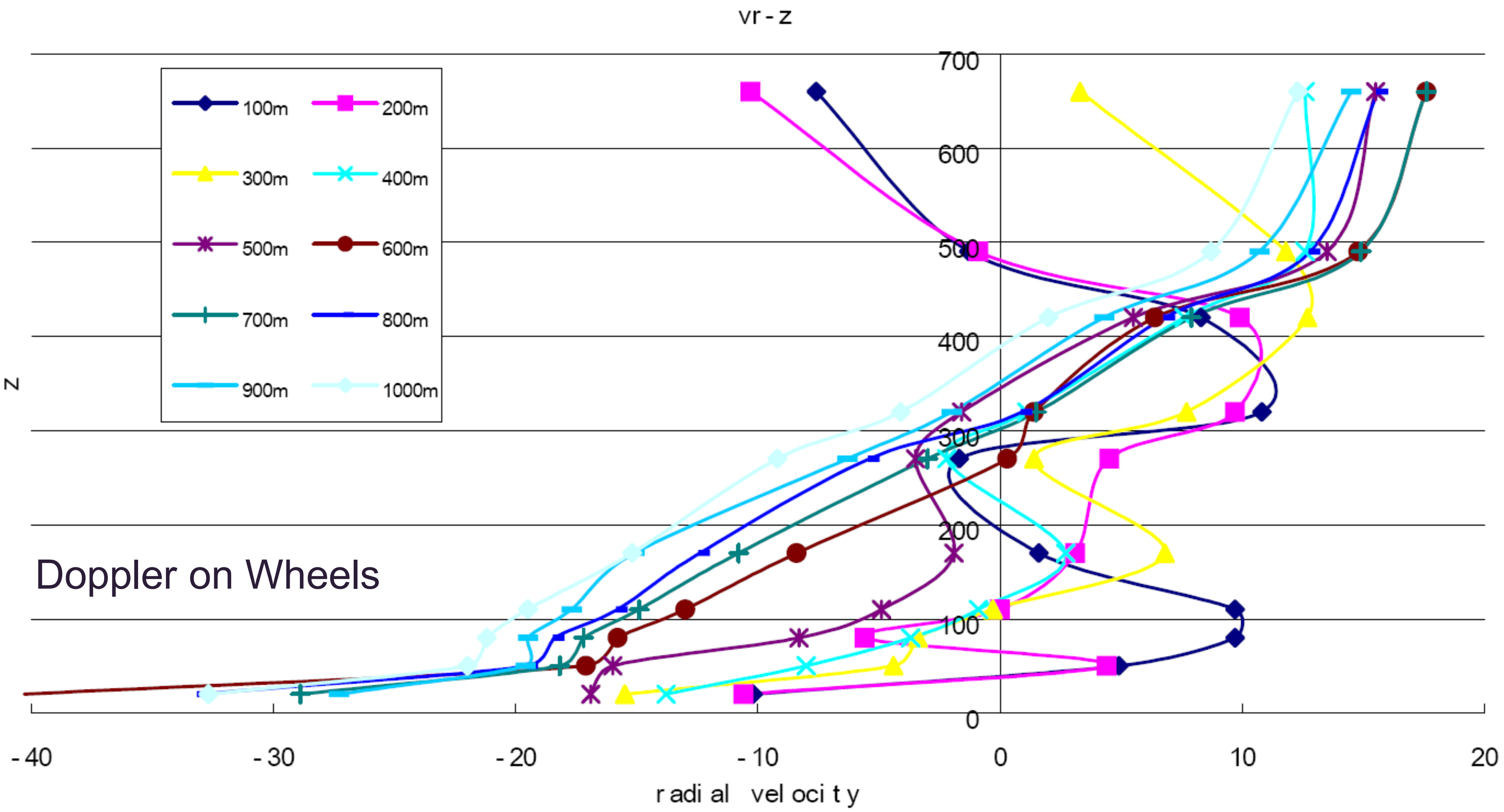



Figure 4: Radar-observed radial velocity profile (m/s) averaged azimuthally as a function of height (in meters). Different curves refer to different radial distances (m) from the center of the vortex.

# Tornado damage

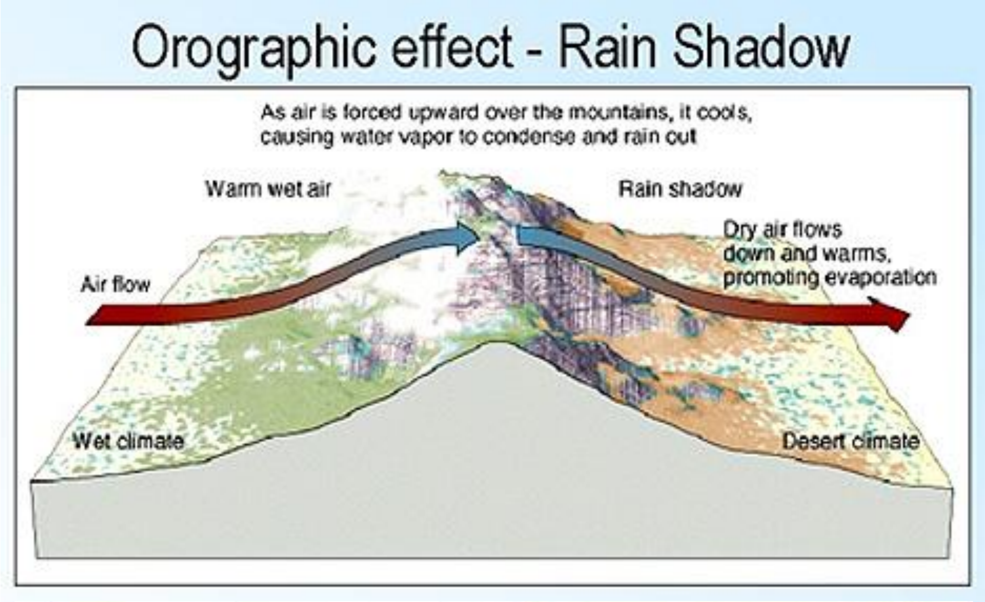

F4 Tornado Moore Oklahoma – Total destruction!!

http://www.spc.noaa.gov/faq/tornado/f4.htm

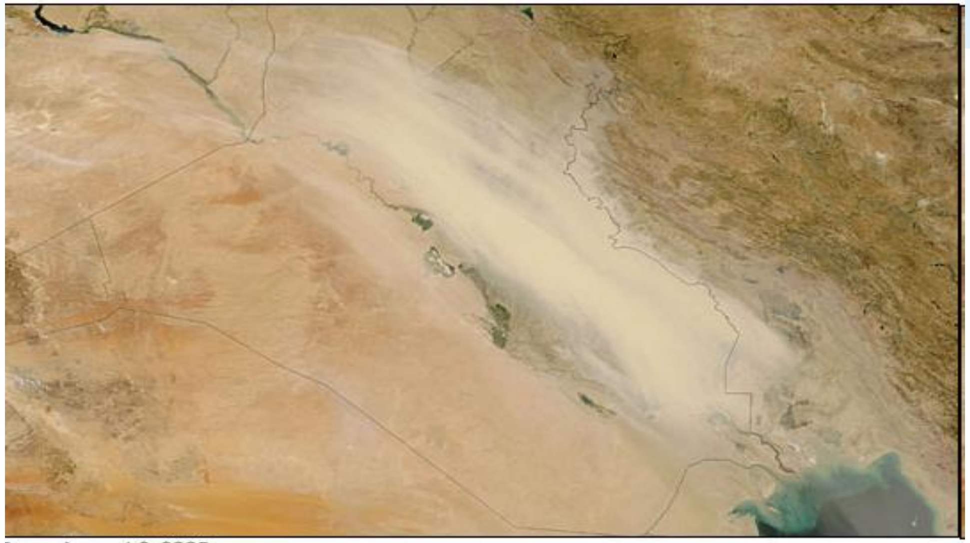
# Other winds

## Orographic winds

- Foehn, Chinook, etc.

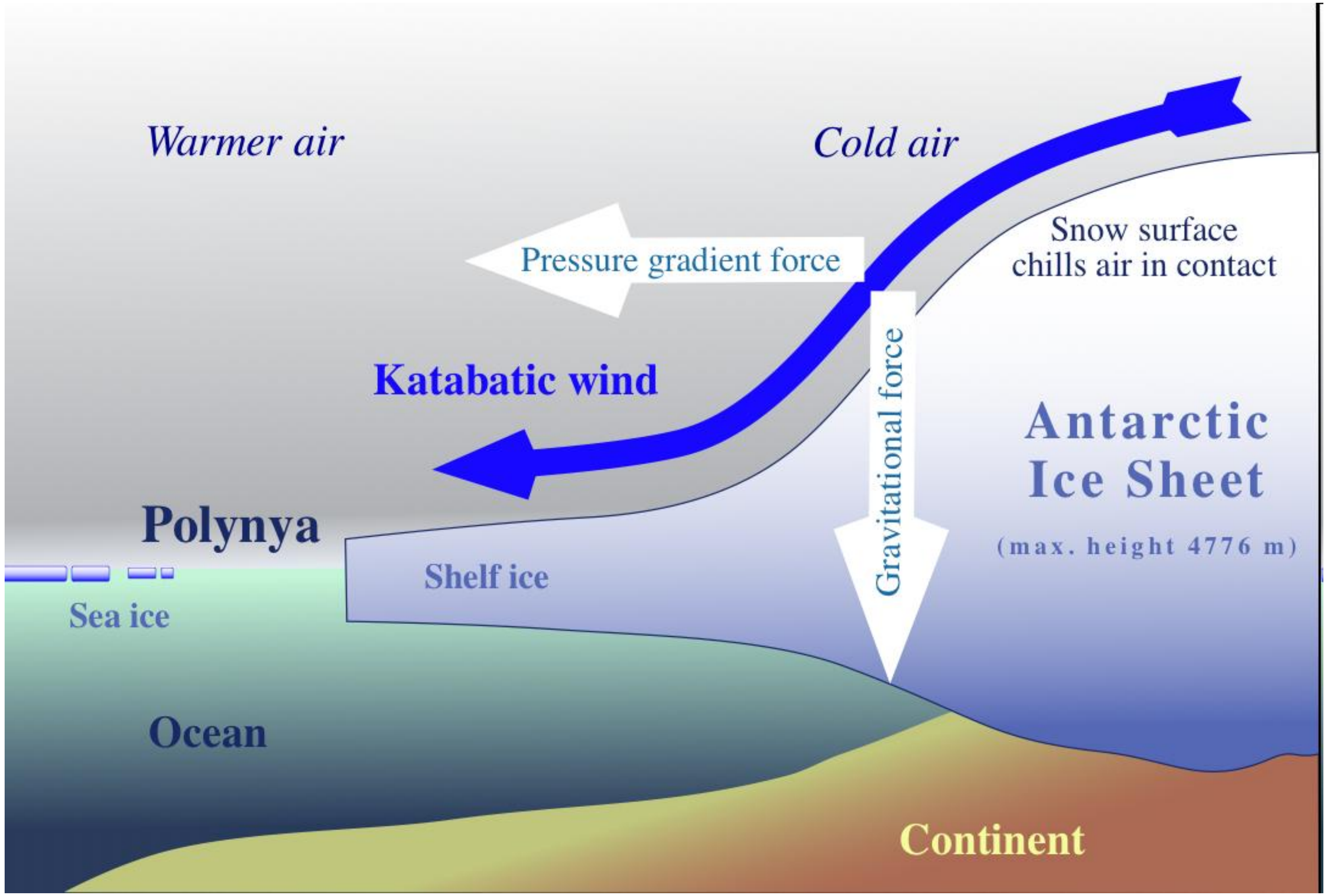



## Shamal winds




http://upload.wikimedia.org/wikipedia/en/c/cd/Iraqdust2005220.jpg

# Other winds




http://upload.wikimedia.org/wikipedia/commons/7/76/Katabatic-wind_hg.png

# Summary

- Many different types of severe windstorms that wind engineers must consider in the wind-resistant design of structures (buildings, wind turbines).
- All have different structure (size, duration, wind profiles) that must be accounted for in this process.
- Problem: most wind-resistant design codes and standards do NOT account for different types of windstorm

# Contact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